H2MWSTNet: A Hybrid Hierarchical Multigranularity Window Shift Transformer Network for Hyperspectral Image Classification

Linshan Zhong, Shuli Cheng, *Member, IEEE*, and Liejun Wang

***Abstract*— Currently, hybrid network models combining convolutional neural networks (CNNs) and Transformers have garnered significant interest among researchers in hyperspectral image (HSI) classification. However, some hybrid models only consider the combination of a single-branch CNN and Transformer, neglecting the shallow multiscale spatial–spectral information and the deeper local spatial information. To address these issues, we propose a hybrid hierarchical multigranularity window shift transformer network (H2MWSTNet), which can efficiently extract multiscale spatial–spectral information and capture local spatial information in sequences. Specifically, we construct a new dual-branch spatial spectral convolution (DBSSC) block. The block effectively realizes the fusion of spatial–spectral information at different scales by aggregating the outputs of both spatial and spectral branches. Furthermore, we have designed a global feature enhancement (GFE) module that effectively enhances the features of the input token sequences. Along with GFE, we propose a multigranularity window shift attention (MWSA) module, which dynamically shifts the key matrix, enhancing the distinguishability of deeply extracted local spatial features. The GFE and MWSA modules provide a novel perspective for HSI feature extraction, constituting a feature mixer based on GFE and multigranularity window shift attention (GMSA) module. Building on the generic architecture and GMSA, we further developed a multigranularity window shift transformer (MWSFormer) encoder block. Finally, our CNN–Transformer hybrid model is designed in a hierarchical fashion, effectively reducing dimensionality and markedly improving classification accuracy. We conducted extensive experiments on six public HSI datasets, and the results demonstrate that H2MWSTNet outperforms the most advanced classification algorithms.**

***Index Terms*— Convolutional neural networks (CNNs), hyperspectral image (HSI), multigranularity, Transformers.**

I. Introduction

HYPERSPECTRAL image (HSI) data are acquired using advanced remote sensing techniques, which provide more detailed and richer spectral information than conventional images by capturing reflectance or radiance data from land cover in dozens or even hundreds of successive bands. Thanks to this advantage, HSI is successfully used in a variety of applications, such as environmental monitoring [1], agricultural surveillance [2], and medical diagnostics [3].

In early HSI classification work, many traditional machine learning techniques were created for HSI classification, such as multinomial logistic regression [4], the K-nearest neighbor algorithm [5], and Bayesian estimation methods [6]. Nonetheless, these conventional approaches depend on manual feature extraction, which is limited in its capability, resulting in lower final classification accuracy.

In recent years, the swift advancement in deep learning technologies has brought unprecedented changes to the field of remote sensing, particularly in the classification of HSIs, which has made significant progress. These technologies include capsule networks [7], recurrent neural networks (RNNs) [8], convolutional neural networks (CNNs) [9], generative adversarial networks (GANs) [10], and graph convolutional networks (GCNs) [11]. Because of its remarkable feature extraction capabilities, CNN has become the mainstream method for processing HSI tasks, ranging from 2D-CNN to hybrid 2D–3D CNN [12], [13], [14]. Furthermore, in order to improve the flexibility in feature selection and enhancement, some researchers have integrated attention mechanisms into CNNs and successfully applied them to HSI classification [15], [16], [17]. For example, Wu et al. [18] proposed a spectral–spatial large kernel attention network (SSLKA) that combines a large convolutional kernel and an attention mechanism. The method effectively extracts local contextual information and captures long-range dependencies through the large kernel attention (LKA) mechanism. Zhao et al. [19] proposed the intermediate domain prototype class learning network (IDPNet), which utilizes multitemporal remote sensing images for segmenting Spartina alterniflora. By dynamically generating intermediate domain features to construct class prototypes, this method effectively addresses the issue of inconsistent representations in remote sensing data across different time periods. Gao et al. [20] proposed GLSANet, which achieves multiscale object detection in remote sensing images through a global–local feature aggregation network. Wang et al. [21] proposed LSCNet, which expands the receptive field through large kernel convolutions and introduces a multifrequency attention (MFA) module to

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replace global average pooling (GAP), enhancing the important information in different feature channels and thereby improving the model's discriminative ability and learning capability.

Although the CNN-based algorithmic models proposed by the aforementioned researchers have shown good performance in spatial–spectral feature extraction, relying solely on a single scale for feature extraction makes it hard to completely harness the spatial–spectral information. Therefore, researchers have developed a series of multiscale dual-branch networks to address this issue [22], [23], [24]. For example, Cheng et al. [25] realized the capture and fusion of spatial–spectral information at different scales by combining CNN and Transformer. Liu et al. [26] proposed a multiarea target attention (MATA) network that can effectively extract spectral–spatial information from target pixels and their similar pixels in a multiscale region through its innovative multiscale target attention module (MSTAM), which significantly improves the accuracy and efficiency of classification. Although the above two-branch structure effectively extracts multiscale spatial–spectral information, methods such as the attention mechanism and GCN are introduced, which may result in the failure to adequately capture all the useful information as well as increase the complexity of the model.

Recently, Transformer-based network models have made a huge impact in the field of natural language processing (NLP). Dosovitskiy et al. [27] introduced a vision transformer (ViT), which for the first time applied the Transformer architecture to computer vision (CV) tasks. It decomposes images into serialized image blocks and inputs them into the Transformer model for processing. Because of robust global modeling capabilities, researchers have developed a series of exploits of the Transformer and successfully applied them to HSI classification tasks [28], [29]. For example, Hong et al. [30] proposed a HSI classification method called SpectralFormer, which utilizes the self-attention technique and cross-layer jump connections to improve the model's processing capability for HSI data. Gao et al. [31] proposed the spatial–spectral perception network (S2PNet) for the classification of multisource remote sensing images. By separately extracting spatial relationships from high-resolution images and spectral relationships from spectral images, this network achieves efficient collaborative classification of multisource data. However, the Transformer approach mentioned above is better suited to capture deep features, but shallow features (e.g., edges, texture, etc.) are also important in HSI classification tasks. CNNs excel in this area, and relying on the Transformer architecture alone may not perform well in capturing these shallow features, leading to performance limitations.

In the field of HSI classification, hybrid architectures based on CNN–Transformer are favored by Sun et al. [32], Ouyang et al. [33], Zhao et al. [34], and Zhang et al. [35]. Mei et al. [36] proposed the group-aware hierarchical transformer (GAHT), which improves attention function by introducing a new grouped convolution block, thereby mitigating feature overdispersion. Sun et al. [37] proposed a MASSFormer network, which utilizes shallow CNNs to extract spectral–spatial features and combines them with the memory-enhanced transformer (MATE) for feature learning to significantly improve the classification performance. In [38], a new morphological transformer (morphFormer) was proposed, which conducts spectral–spatial morphological convolutions using erosion and dilation operators and embeds them into a new attention mechanism to effectively fuse morphological features. Wang et al. [39] proposed a multistage information complementary fusion network based on flexible mixing, known as Flex-MCFNet. This network employs a flexible data augmentation strategy to achieve efficient classification of HSI and other modalities, such as multispectral and synthetic aperture radar images, through the interaction and fusion of information at multiple stages. Although the previously mentioned hybrid network models of CNN and Transformer achieved good classification results, these hybrid models did not consider the multiscale spatial–spectral information in the shallow layers, nor did they focus on the local spatial information in the deep layers. In order to intuitively illustrate the focus issue and the role of this information, we adopt the t-SNE method to visualize the high-dimensional output features of some hybrid models on the Salinas dataset, as shown in Fig. 1, where different colors indicate different classes. For areas with noticeable category confusion (the purple and green regions), we used ellipses to mark the misclassified pixels. It can be observed that the GAHT method shows the most dispersed category distribution, with noticeable category confusion. In contrast, the morphFormer method achieves improved intraclass compactness, though some features still overlap. The MVAHN method demonstrates further concentration of category distributions, with clearer category boundaries, yet slight overlaps remain. Our method effectively integrates shallow multiscale spatial–spectral information with deep local spatial information, resulting in the clearest category distribution with well-defined boundaries and almost no noticeable category confusion, outperforming other methods.

Building on the analysis above, we propose a hybrid hierarchical multigranularity window shift transformer network (H2MWSTNet) for HSI classification. First, we design a dual-branch spatial spectral convolution (DBSSC) block to extract the spatial and spectral information of HSI separately, which is able to effectively fuse the spatial–spectral information of different scales and make the output shallow features richer. Additionally, we have designed a global feature enhancement (GFE)-based multigranularity window shift attention (GMSA) module. This module facilitates the interaction among local tokens by dynamically shifting keys in the feature map, significantly enhancing the relevance of local spatial features. Finally, we employ a hierarchical CNN–Transformer hybrid network model. Compared to other hybrid models, the hierarchical model progressively reduces the number of channels in the feature maps, extracting multiscale features in the gradually contracting method, which not only lowers computational costs but also reduces redundant information.

The main contributions of this article are as follows.

1) In order to facilitate the interaction of different scale features, we designed a new DBSSC block. This convolution block is divided into two parts, extracting

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spatial position features and spectral sequence features, respectively, and then the information from these two parts is summed up by feature fusion and input into the subsequent multigranularity window shift transformer (MWSFormer) encoder block. This block effectively fuses the multiscale spatial–spectral information and enhances the feature expression capability.

2) While some existing multihead attention modules excel at extracting global information, they may overlook local details, which are crucial for HSI. In order to overcome these problems, we introduce a new GMSA module composed of the GFE module and the multigranularity window shift attention (MWSA) module. Compared to the existing methods that focus on local spatial information such as CSWin Transformer [40] and spectral Swin Transformer (SSWinT) [41], GMSA fully exploits the rich information in the keys in the input section to enhance the global features. Additionally, GMSA facilitates the interaction among local tokens by dynamically shifting keys in the feature map, significantly enhancing the relevance of local spatial features.
3) To reduce information redundancy, we designed a hierarchical structure. By modeling the DBSSC block and the MWSFormer encoder block in a continuous hierarchical fashion, H2MWSTNet can effectively utilize multilevel information to enhance the representativeness and robustness of features. While reducing the number of channels, it also maintains competitive classification results.

The rest of this article is organized as follows. Section II describes the related work in this article. In Section III, our proposed H2MWSTNet is described in detail. Section IV performs a number of comparative and ablation experiments on four HSI datasets to validate the effectiveness of the proposed method. Finally, the article is summarized in Section V.

II. Related Work

This section is divided into three parts to briefly summarize the related work, namely: a CNN-based method for HSI classification, a Transformer-based method for HSI classification, and a CNN and Transformer-based method for HSI classification.

A. CNN-Based Method for HSI Classification

CNNs, due to their excellent characteristics such as local perception, parameter sharing, and translation invariance, are widely applied in HSI classification and have achieved commendable classification results. Hu et al. [9] proposed a simple yet effective 1D-CNN network, which consists of a five-layer structure with each layer operating on spectral features, effectively extracting spectral features. However, 1D-CNN overlooks the rich spatial information in HSIs. Therefore, Cai et al. [12] proposed a novel 2D-CNN network that uses different convolutional kernels to extract information from various spatial locations and employs penalization and reward strategies to enhance features. However, when considering the spatial and spectral information independently in HSIs, the classification results are not satisfactory, failing to fully exploit the rich 3-D structural features of HSIs and overlooking the combined features of spatial–spectral information. To address these issues, Qi et al. [42] proposed a 3D-CNN approach that uses 3D-CNN to extract and jointly model the spatial–spectral information in HSI, significantly improving classification accuracy. However, compared to 2D-CNN, 3D-CNN has a higher computational complexity. Furthermore, the convolution operations of 3D-CNN may not be sufficiently flexible for extracting complex spectral information in hyperspectral data, as they primarily focus on spatiotemporal relationships, which can result in insufficient expressive capability for spectral features. Therefore, Ma et al. [43] introduced a hybrid 3D- and 2D-CNN approach that incorporates depthwise separable convolutions to design a lightweight network model. This model not only better integrates spatial and spectral information but also significantly reduces computational costs. For multivision processing of HSI classification task, Liu et al. [44] proposed a heterogeneous dual enhancement network (HDECGCN) to extract complementary spatial spectral information by augmenting the CNN and GCN subnetworks to enhance classification precision. Furthermore, Liu et al. [45] proposed a spectral drift suppression (SSM) method to improve the performance of cross-scene HSI classification by reducing amplitude offsets and spectral variations. Although the CNN-based networks have good results in feature extraction, it is difficult to fully utilize the spatial–spectral information by relying only on a single scale to extract features. Moreover, due to inherent architectural limitations, CNN can only extract shallow, local spatial–spectral information and is incapable of capturing long-range dependencies. Therefore, some researchers have begun to focus on HSI classification tasks based on ViT.

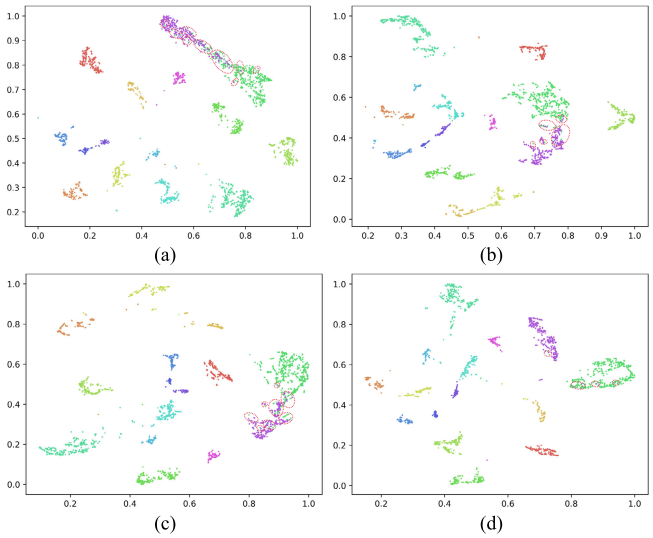


Fig. 1. t-SNE visualization of the output features of some hybrid models on the Salinas dataset. (a) GAHT. (b) MVAHN. (c) morphFormer. (d) H2MWSTNet.

B. Transformer-Based Method for HSI Classification

In recent years, with the exceptional performance of Transformers in the fields of NLP and CV, researchers have

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begun to introduce them to the task of HSI classification to explore their potential in modeling spatial–spectral information and HSI classification. Unlike CNN, which uses static filters to extract local information, ViT captures long-distance dependencies and remote interactions in multihead self-attention in a dynamic manner through parallel computation. Hong et al. [30] proposed a novel Transformer-based network that captures local sequential information of adjacent bands through group spectral embeddings and then uses cross-layer skip connections to feed shallow information into the Transformer, which helps to reduce information loss. He et al. [46] proposed a Transformer model that utilizes a spectral transformer (Spe-TR) and a spatial transformer (Spa-TR) to separately extract spectral and spatial features. The features from these two branches are then fused, effectively enhancing the accuracy of land cover classification. Liao et al. [47] used a Transformer framework to construct a spatial–spectral Transformer module, respectively, to capture spectral and spatial long-range dependencies in HSI, and then effectively fused the obtained high-level semantic features through an adaptive spatial–spectral fusion mechanism. Feng et al. [48] proposed a spectral–spatial-elevation fusion transformer (S2EFT) that utilizes a sliding-grouped spectral embedding module and a spatial information recognition module to separately extract spectral and spatial information. Subsequently, an attention mechanism is used to fuse these pieces of information, effectively enhancing the transmission of spectral–spatial data. Jiang et al. [49] proposed a graph generation structure-aware transformer (GraphGST) that incorporates absolute position encoding and a self-supervised learning generation mechanism to facilitate cross-view contrastive learning, thereby enhancing the representation learning capabilities. Although the performance of the aforementioned Transformer methods is satisfactory, they still fall short in capturing local features that are crucial for HSI classification. Therefore, relying solely on Transformers is insufficient for handling complex HSI classification tasks.

C. CNN and Transformer-Based Method for HSI Classification

Combining CNNs with Transformers leverages the maximum strengths of both, making this hybrid approach popular among many researchers [50]. To fully exploit their complementary advantages, Zhao et al. [51] proposed a grouped separable convolutional ViT (GSC-ViT), which utilizes grouped separable convolution (GSC) modules and grouped separable multihead self-attention (GSSA) modules, effectively reducing the number of parameters and enhancing the feature extraction capacity. Zhou et al. [52] also proposed a dual-branch convolutional transformer network, named DCTN. It employs 3-D convolutions and 2-D grouped convolutions to simultaneously extract rich spatial–spectral information and uses the efficient interactive self-attention (EISA) mechanism to fuse information from its height–width-spectral dimensions. Zhang et al. [53] proposed a multilevel DA network (MDA-NET) for cross-scene HSI classification through multilevel domain alignment and cross-modal joint classification. Ye et al. [54] proposed the adaptive-domain adversarial few samples learning (ADAFSL) method, which significantly improves the classification performance of cross-domain HSIs by combining an adaptive-domain adversarial strategy and a multiscale feature extraction module. Mohamed et al. [55] proposed a Transformer architecture for factorization called FactoFormer. It utilizes a factorization architecture to enable models to learn and fuse independently in spectral and spatial dimensions to more accurately capture the complexity of hyperspectral data, which is then pretrained by masked image modeling (MIM) techniques, which greatly improves the model's ability to generalize over limited labeled data. Shi et al. [56] proposed the multiscale hierarchical conv-aided FourierFormer (MHCFormer), which combines Fourier transform, Transformer, and CNN for HSI classification. It captures the complex relationships between spectral and spatial dimensions through multiscale spectral–spatial tagging and hierarchical tagging learning. This shows that the joint CNN and Transformer modeling approach is sought after. However, it remains challenging to fully exploit the multiscale spatial–spectral information in CNN and the local spatial information in Transformer.

III. Proposed Method

A. Proposed Overall Structure

Our designed H2MWSTNet framework is illustrated in Fig. 2, which consists of four parts: DBSSC, MWSFormer, GAP, and a classifier. Specifically, a three-stage hierarchical framework was designed, progressively reducing the number of output channels and modeling low, medium, and high-level features, respectively, thereby extracting multilevel information. Multilevel information refers to the information obtained from different feature extraction layers. Each stage consists of a DBSSC block and several MWSFormer blocks, where the DBSSC is utilized to extract shallow local multiscale spatial and spectral information, which is then fed into the MWSFormer blocks to capture deep global spectral and local spatial multigranularity information. Multiscale information refers to the combination of spectral features at different spatial scales to capture more comprehensive spectral–spatial information. Multigranularity information refers to the extraction and representation of features at different granularities within HSI. This local–global strategy enables efficient modeling of the spatial–spectral feature in HSI.

In this article, we take an input HSI patch cube $X \in \mathbb{R}^{P\times P\times C}$, where P is the input HSI patch and C represents the HSI bands. We supply the HSI patch to a three-stage sequence composed of DBSSC and MWSFormer, which learn different levels of spatial–spectral features in the HSI, respectively. The feature map sizes corresponding to these three stages are as follows: $P \times P \times C_1$, $P \times P \times C_2$, and $P \times P \times C_3$. The number of MWSFormer blocks at each level is denoted by n_1, n_2, and n_3, respectively. Next, we use GAP to compress the spatial information in the last level of output and feed it into the classifier.

B. Dual-Branch Spatial–Spectral Convolution Block

Most of the patch embedding modules in ViT use a single scale to extract features, which hinders the full utilization

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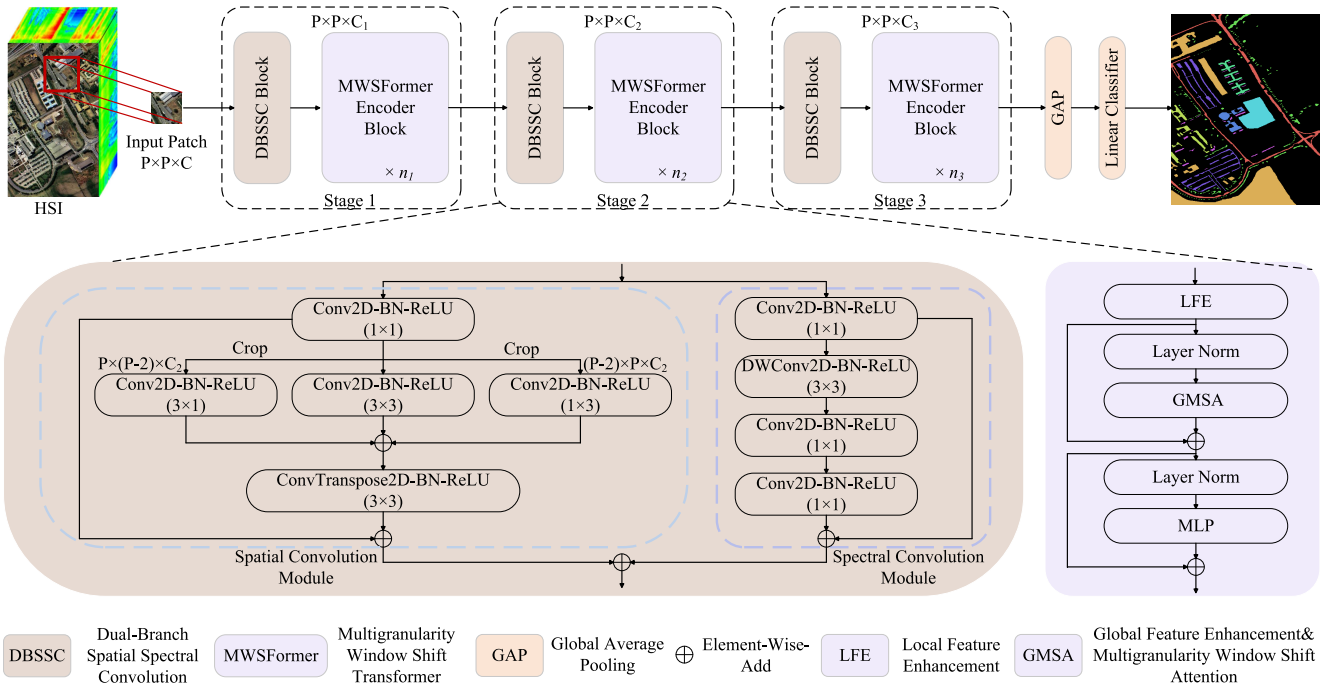


Fig. 2. Proposed H2MWSTNet network architecture. The network is divided into three phases, each with a DBSSC block and several MWSFormer encoder blocks.

of spatial–spectral information, thus limiting the room for improvement in classification performance. To obtain richer feature information, this article proposes a DBSSC block consisting of a SpaC module and a spectral convolution (SpeC) module. They are responsible for extracting spatial location features and spectral sequence features, respectively, by fusing the features from these two modules. Finally, we embed the output of DBSSC into the MWSFormer encoder block to further improve the classification performance.

1) SpaC Module: To effectively extract local semantic information from HSI, we extract features from both spatial and spectral perspectives. In the SpaC module, we use a five-layer convolution block to extract shallow spatial features. After each convolutional layer, there is a batch normalization (BN), followed by a rectified linear unit (ReLU). Specifically, given an HSI feature map of size $\boldsymbol{X} \in \mathbb{R}^{C\times H\times W}$, where H and W are the height and width of the feature map, respectively, and C is the number of channels, we use convolutions with varying kernel sizes to extract features from different spatial locations. Notably, to accommodate the kernel size, we cropped the input features of the vertical and horizontal convolution blocks. This reduces the boundary effect on hyperspectral feature extraction and ensures the effectiveness of the convolution operations. The result $\boldsymbol{T}_1$ is shown in the following equation:

$$\boldsymbol{T}_1 = \mathrm{SpaC}(\boldsymbol{X}) \in \mathbb{R}^{C\times H\times W}. \tag{1}$$

2) SpeC Module: The SpeC module is one branch of the DBSSC, parallel to the SpaC module. It utilizes a four-layer convolution block to extract shallow spectral information. Specifically, a 2-D convolution with a kernel of 1×1 is used to map the channels of the feature map to specified locations for subsequent computation and processing. Depth-separable convolution is then used to separate the spatial and spectral features, preserving information about the channel distribution. Similarly, each convolutional layer is followed by a BN and a ReLU, respectively. The SpaC module and the SpeC module share the same input, which is the HSI feature map of $\boldsymbol{X} \in \mathbb{R}^{C\times H\times W}$. The result $\boldsymbol{T}_2$ is shown in the following equation:

$$\boldsymbol{T}_2 = \mathrm{SpeC}(\boldsymbol{X}) \in \mathbb{R}^{C\times H\times W}. \tag{2}$$

Finally, by adding these two modules together, the local spatial and spectral features are effectively integrated, enhancing the correlation between spatial and spectral information and improving the H2MWSTNet's ability to represent multiscale features in complex HSIs. The result $\boldsymbol{T}$ is shown in the following equation:

$$\boldsymbol{T} = \mathrm{SpaC}(\boldsymbol{X}) + \mathrm{SpeC}(\boldsymbol{X}) \in \mathbb{R}^{C\times H\times W}. \tag{3}$$

C. MWSFormer Encoder Block

The MWSFormer encoder block is the core of the H2MWSTNet model, used for learning local and global information from the input tokens. As shown in Fig. 2, the MWSFormer encoder block mainly consists of local feature enhancement (LFE), GMSA, and multilayer perceptron (MLP). First, local information representation is enhanced via the convolutional architecture of LFE. Furthermore, there is a layer of normalization (LN) before and after the GMSA, and residual connections are placed after each LN to mitigate the problem of vanishing or exploding gradients.

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1) Local Feature Enhancement: To further extract local features from HSI, we propose an LFE module, which includes a depthwise separable convolution and two 2-D convolutions. Then, we use LFE to further extract the local spatial–spectral information of HSI, and its output is consistent with the output of DBSSC. Introducing the LFE module can increase the model's nonlinear representation capability, enabling it to adapt to complex data distributions and decision surfaces. This enhances the model's generalization capability and overall performance in hyperspectral classification tasks and better prepares it for the subsequent GMSA.

2) Multigranularity Window Shift Attention: To enhance the sensitivity of different positional sequences, we propose a novel local attention module for GMSA. Unlike other attention modules, we first fully explore the wealth of information in the keys in the input part to enhance global features, and then we use a multigranularity window shift method for local attention calculation to extract local spatial information. Here, multigranularity refers to the use of multiple different values for the window size (i.e., granularity). Fig. 3 shows the overall framework of the GMSA. Specifically, we produce one 2-D input matrix $\boldsymbol{X} \in \mathbb{R}^{n\times d}$ by compressing the output spatial dimensions of the LFE module and use three weight matrices $\boldsymbol{W}^q$, $\boldsymbol{W}^k$, and $\boldsymbol{W}^v$ to linearly map the compressed token sequence $\boldsymbol{X} \in \mathbb{R}^{n\times d}$ of the LFE subnetwork into query ($\boldsymbol{Q}$), key ($\boldsymbol{K}$), and value ($\boldsymbol{V}$). Wherein, $\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V} \in \mathbb{R}^{n\times d}$ and $\boldsymbol{W}^q, \boldsymbol{W}^k, \boldsymbol{W}^v \in \mathbb{R}^{d\times d}$, where n is the square of the HSI patch size and d is the number of channels. The specific formula is shown in the following equation:

$$\boldsymbol{Q} = \boldsymbol{X}\boldsymbol{W}^q, \quad \boldsymbol{K} = \boldsymbol{X}\boldsymbol{W}^k, \quad \boldsymbol{V} = \boldsymbol{X}\boldsymbol{W}^v. \tag{4}$$

Next, we multiply the key matrix $\boldsymbol{K}$ by the learnable parameter vector $\boldsymbol{W}^\alpha \in \mathbb{R}^d$ to learn the attention weights of the keys, obtaining the global attention key vector α. By learning the parameter vector, attention weights can be adaptively adjusted to accommodate differences between samples or correlations among different features. Thus, with enhanced flexibility, it can better adapt to the specific feature distribution and correlations when dealing with different HSIs. The specific formula is shown in the following equation:

$$\boldsymbol{\alpha} = \boldsymbol{K} \cdot \boldsymbol{W}^\alpha / \sqrt{d} \tag{5}$$

where $\boldsymbol{\alpha} \in \mathbb{R}^n$ is the global attention key vector, $\sqrt{d}$ is the scaling factor used to adjust the attention weights, and finally, we normalize the vector $\boldsymbol{\alpha}$ along the channel dimension.

Then, based on the learned attention weights, perform elementwise multiplication on the key matrix and sum them along the vector dimension. The specific formula is shown in the following equation:

$$\boldsymbol{K}' = \sum_{i=1}^{n} \boldsymbol{\alpha}_i * \boldsymbol{K}_i \tag{6}$$

where $\boldsymbol{K}' \in \mathbb{R}^d$ is the single global key vector, $\boldsymbol{\alpha}_i$ represents the ith key vector, $\boldsymbol{K}_i$ represents the ith key matrix, and $*$ denotes the elementwise multiplication operation.

Next, we perform a replication operation on the global key vector $\boldsymbol{K}'$. Specifically, we repeat along the channel dimension, using the number of tokens in the query or value as the number of repetitions, allowing the global key vector to form a global context ($\boldsymbol{K}' \in \mathbb{R}^{n\times d}$). This matrix has similarities to traditional attention matrices that capture information about each token and can be a good representation of correlations in a sequence.

To enhance the global visual representation capability, we use the dynamically learned new key matrix ($\boldsymbol{K}'$), which is concatenated with the original query matrix ($\boldsymbol{Q}$) and value matrix ($\boldsymbol{V}$) along the spectral dimension, respectively. This allows the model to better utilize the characteristics of the input data, hence improving the classification model's understanding and predictive performance on hyperspectral data. Such operations help our model better differentiate the features across various bands and enhance its understanding of the relationships between different wavelengths, contributing to a more precise categorization of various classes within hyperspectral data. The specific formulas are shown in the following equations:

$$\boldsymbol{Q}' = \text{Concat}\left(\boldsymbol{Q}, \boldsymbol{K}'\right) \tag{7}$$

$$\boldsymbol{V}' = \text{Concat}\left(\boldsymbol{V}, \boldsymbol{K}'\right) \tag{8}$$

where $\boldsymbol{Q}', \boldsymbol{V}' \in \mathbb{R}^{n\times 2d}$ are the new query and value matrices, respectively, enhanced through feature augmentation, $\boldsymbol{K}' \in \mathbb{R}^{n\times d}$ is the new key matrix dynamically learned and obtained through a series of operations, and $\boldsymbol{Q}, \boldsymbol{V} \in \mathbb{R}^{n\times d}$ are the original query and value matrices.

To facilitate subsequent matrix calculations and correspond to the outputs of different layers, we use a fully connected layer to linearly project the dimensions of $\boldsymbol{Q}'$ and $\boldsymbol{V}'$ to the same dimensions as $\boldsymbol{K}'$. The specific formulas are shown in the following equations:

$$\boldsymbol{Q}' = \text{Linear}\left(\boldsymbol{Q}'\right) \in \mathbb{R}^{n\times d} \tag{9}$$

$$\boldsymbol{V}' = \text{Linear}\left(\boldsymbol{V}'\right) \in \mathbb{R}^{n\times d}. \tag{10}$$

Although possessing powerful global modeling capabilities, the original complete attention mechanism excessively focuses on the spectral dimension, leading to insufficient utilization of local spatial information, which plays a critical role in recognizing ground object categories. To alleviate this issue, we propose an MWSA mechanism, which dynamically shifts the keys to facilitate local token interactions, followed by attention calculations within the multigranularity window, as shown in Fig. 3.

After GFE, the dimensions of $\boldsymbol{Q}'$, $\boldsymbol{K}'$, and $\boldsymbol{V}'$ are reshaped from $n \times d$ to $h \times w \times d$, where $n = h \times w$. In addition, we linearly project the input features to g attention heads, and then each head performs local self-attention within multiple granularity windows. In the local self-attention process, $\boldsymbol{Q}'$, $\boldsymbol{K}'$, and $\boldsymbol{V}'$ are evenly partitioned into m nonoverlapping granularity windows of width sh, with each granularity window sized $h \times sh$. Where h is the length of the granularity window and sh is the width of the granularity window, their feature learning capabilities and computational complexity can be adjusted by controlling their length and width. The specific

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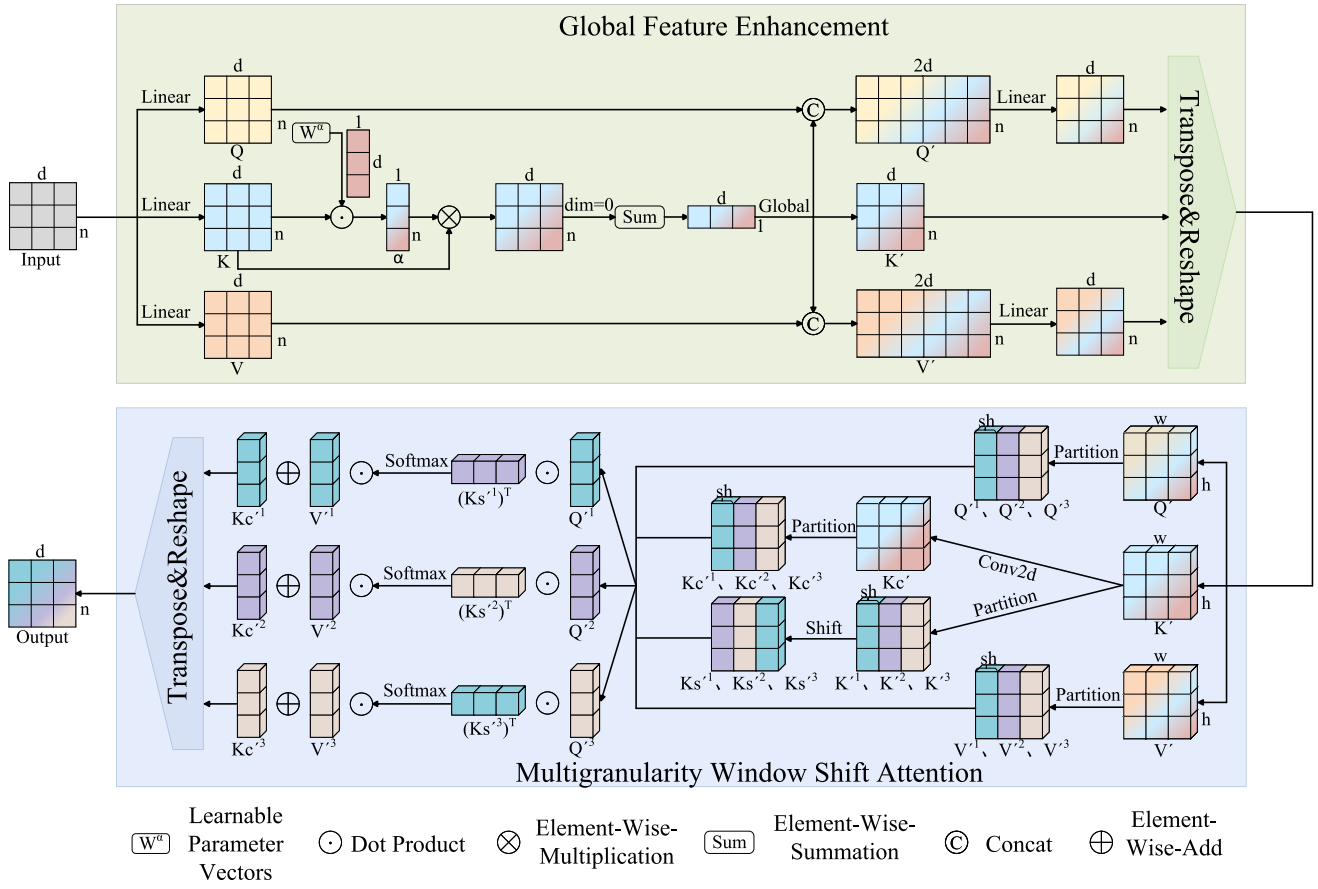


Fig. 3. Proposed GMSA structure. It consists of two parts: GFE and MWSA.

formulas are shown in the following equations:

$$\begin{aligned} \boldsymbol{Q}' &= \left[\boldsymbol{Q}'^1, \boldsymbol{Q}'^2, \ldots, \boldsymbol{Q}'^i, \ldots, \boldsymbol{Q}'^m\right] \\ \boldsymbol{K}' &= \left[\boldsymbol{K}'^1, \boldsymbol{K}'^2, \ldots, \boldsymbol{K}'^i, \ldots, \boldsymbol{K}'^m\right] \\ \boldsymbol{V}' &= \left[\boldsymbol{V}'^1, \boldsymbol{V}'^2, \ldots, \boldsymbol{V}'^i, \ldots, \boldsymbol{V}'^m\right] \end{aligned} \tag{11}$$

$$\begin{aligned} \boldsymbol{Q}'^i &= \left[\boldsymbol{Q}'^i_1, \boldsymbol{Q}'^i_2, \ldots, \boldsymbol{Q}'^i_e, \ldots, \boldsymbol{Q}'^i_g\right] \\ \boldsymbol{K}'^i &= \left[\boldsymbol{K}'^i_1, \boldsymbol{K}'^i_2, \ldots, \boldsymbol{K}'^i_e, \ldots, \boldsymbol{K}'^i_g\right] \\ \boldsymbol{V}'^i &= \left[\boldsymbol{V}'^i_1, \boldsymbol{V}'^i_2, \ldots, \boldsymbol{V}'^i_e, \ldots, \boldsymbol{V}'^i_g\right] \end{aligned} \tag{12}$$

where $\boldsymbol{Q}', \boldsymbol{K}', \boldsymbol{V}' \in \mathbb{R}^{n\times d}$ and have already been partitioned into m granularity windows. $\boldsymbol{Q}'^i, \boldsymbol{K}'^i, \boldsymbol{V}'^i \in \mathbb{R}^{(h\times sh)\times d}$ represent their states under the ith granularity window, where their number of attention heads is linearly projected onto g heads. $\boldsymbol{Q}'^i_e, \boldsymbol{K}'^i_e, \boldsymbol{V}'^i_e \in \mathbb{R}^{(h\times sh)\times (d/g)}$ represent the query, key, and value matrices under the ith granularity window and the eth head, respectively. $m = w/sh$ and $i = 1, \ldots, m$. Next, the multihead key matrices are shifted. The shifting operation can adjust the positions of the key matrices, making the attention weights at each position more accurately correspond to the surrounding features, thus enhancing the model's perception of local features and improving its understanding and representation of data. Assuming the number of attention heads is e, the specific formula is shown in the following equations:

$$\boldsymbol{Ks}' = \left[\boldsymbol{K}'^2, \boldsymbol{K}'^3, \ldots, \boldsymbol{K}'^i, \ldots, \boldsymbol{K}'^m, \boldsymbol{K}'^1\right] \tag{13}$$

$$\boldsymbol{Ks}'^i = \left[\boldsymbol{K}'^i_1, \boldsymbol{K}'^i_2, \ldots, \boldsymbol{K}'^i_e, \ldots, \boldsymbol{K}'^i_{g-1}, \boldsymbol{K}'^i_g\right] \tag{14}$$

where $\boldsymbol{Ks}' \in \mathbb{R}^{n\times d}$ is the new key matrix after performing the dynamic shift, and m is the number of granularity windows for the partition. $\boldsymbol{Ks}'^i \in \mathbb{R}^{(h\times sh)\times d}$ represents the shift key matrix under the ith granularity window. Finally, the attention score of the eth head is calculated by scaling the Softmax function. The specific formulas are shown in the following equations:

$$\boldsymbol{Z}^i_e = A\left(\boldsymbol{Q}'^i_e, \boldsymbol{Ks}'^i_e, \boldsymbol{V}'^i_e\right) + \mathrm{LEPE}\left(\boldsymbol{K}'\right) \tag{15}$$

$$A\left(\boldsymbol{Q}'^i_e, \boldsymbol{Ks}'^i_e, \boldsymbol{V}'^i_e\right) = F\left(\frac{\boldsymbol{Q}'^i_e\left(\boldsymbol{Ks}'^i_e\right)^T}{\sqrt{d}}\right)\boldsymbol{V}'^i_e \tag{16}$$

$$\boldsymbol{Z}_e = \left[\boldsymbol{Z}^1_e, \boldsymbol{Z}^2_e, \ldots, \boldsymbol{Z}^m_e\right] \tag{17}$$

where $\boldsymbol{Q}'^i_e, \boldsymbol{Ks}'^i_e, \boldsymbol{V}'^i_e \in \mathbb{R}^{(h\times sh)\times (d/g)}$ respectively represent the query, shifted key, and value matrices at the ith granularity window and the eth head. $\mathrm{LEPE}(\boldsymbol{K}')$ represents local-enhanced positional encoding, which employs a 2-D convolution with a kernel of 3×3 to process key matrices containing rich contextual information. This approach introduces more informative feature representations for each position, followed by the partition into granularity windows, as previously mentioned. These encodings help the model

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TABLE I
LAND COVER CATEGORIES OF THE SA DATASET AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CATEGORY

Class No.	Class name	Train	Val	Test
1	Brocoli_green_weeds_1	20	20	1969
2	Brocoli_green_weeds_2	37	37	3652
3	Fallow	20	20	1936
4	Fallow_rough_plow	14	14	1366
5	Fallow_smooth	27	27	2624
6	Stubble	39	40	3880
7	Celery	36	36	3507
8	Grapes_untrained	113	112	11046
9	Soil_vinyard_develop	62	62	6079
10	Corn_senesced_green_weeds	33	33	3212
11	Lettuce_romaine_4wk	11	10	1047
12	Lettuce_romaine_5wk	19	20	1888
13	Lettuce_romaine_6wk	9	9	898
14	Lettuce_romaine_7wk	10	11	1049
15	Vinyard_untrained	73	72	7123
16	Vinyard_vertical_trellis	18	18	1771
	Total	541	541	53047

TABLE II
LAND COVER CATEGORIES OF THE PU DATASET AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CATEGORY

Class No.	Class name	Train	Val	Test
1	Asphalt	66	66	6499
2	Meadows	186	187	18276
3	Gravel	21	21	2057
4	Trees	30	31	3003
5	Painted metal sheets	14	13	1318
6	Bare Soil	50	50	4929
7	Bitumen	13	14	1303
8	Self-Blocking Bricks	37	37	3608
9	Shadows	10	9	928
	Total	427	428	41921

TABLE III
LAND COVER CATEGORIES OF THE WHU-LK DATASET AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CATEGORY

Class No.	Class name	Train	Val	Test
1	Corn	172	173	34166
2	Cotton	42	42	8290
3	Sesame	15	15	3001
4	Broad-leaf soybean	316	316	62580
5	Narrow-leaf soybean	21	21	4109
6	Rice	59	60	11735
7	Water	335	335	66386
8	Roads and houses	36	35	7053
9	Mixed weed	26	26	5177
	Total	1022	1023	202497

TABLE IV
LAND COVER CATEGORIES OF THE HR-L DATASET AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CATEGORY

Class No.	Class name	Train	Val	Test
1	Dense Urban Fabric	8	9	271
2	Mineral Extraction Sites	2	2	63
3	Non Irrigated Arable Land	17	16	509
4	Fruit Trees	3	2	74
5	Olive Groves	42	42	1317
6	Broad-leaved Forest	7	6	210
7	Coniferous Forest	15	15	470
8	Mixed Forest	32	32	1008
9	Dense Sclerophyllous Vegetation	114	114	3565
10	Sparce Sclerophyllous Vegetation	84	84	2635
11	Sparcely Vegetated Areas	12	12	380
12	Rocks and Sand	14	15	458
13	Water	42	42	1309
14	Coastal Water	13	14	424
	Total	405	405	12693

TABLE V
LAND COVER CATEGORIES OF THE IP DATASET AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CATEGORY

Class No.	Class name	Train	Val	Test
1	Alfalfa	2	3	41
2	Corn-notill	71	72	1285
3	Corn-mintill	42	41	747
4	Corn	12	12	213
5	Grass-pasture	24	24	435
6	Grass-trees	37	36	657
7	Grass-pasture-mowed	2	1	25
8	Hay-windrowed	24	24	430
9	Oats	1	1	18
10	Soybean-notill	49	48	875
11	Soybean-mintill	122	123	2210
12	Soybean-clean	30	29	534
13	Wheat	10	10	185
14	Woods	63	63	1139
15	Buildings-Grass-Trees-Drives	19	20	347
16	Stone-Steel-Towers	4	5	84
	Total	512	512	9225

better understand the significance and context of each position, thus enabling a more accurate calculation of attention scores. $\boldsymbol{Z}_e^i \in \mathbb{R}^{(h\times sh)\times(d/g)}$ represents the attention scores under the ith granularity window and the eth head. $\sqrt{d}$ is used to scale the attention weights to prevent influence from the number of frequency bands. F represents the Softmax activation function, which gives the probability distribution of different pixel positions by normalizing the attention weights. $\boldsymbol{Z}_e \in \mathbb{R}^{n\times(d/g)}$ represents the attention scores of the eth head. Finally, the outputs of multiple heads are concatenated to form the final output of the multihead attention. The specific formula is shown in the following equation:

$$\boldsymbol{Output} = \text{Concat}\big(\boldsymbol{Z}_1, \boldsymbol{Z}_2, \ldots, \boldsymbol{Z}_e, \ldots, \boldsymbol{Z}_g\big)\boldsymbol{W}^O \tag{18}$$

where $\boldsymbol{W}^O \in \mathbb{R}^{d\times d}$ is the commonly used projection matrix that projects the results of the multihead attention into the target output dimension (default set to d). $\boldsymbol{Output} \in \mathbb{R}^{n\times d}$ represents the final attention score. Since our network is divided into three phases, the number of MWSFormer blocks in each stage varies, which will be discussed in Section IV.

IV. EXPERIMENTS

In this section, we utilize four benchmark datasets to validate the effectiveness of the proposed method. Subsequently, we conduct an ablation analysis on the proposed module to qualitatively assess the HSI classification performance of our method. Finally, we compare our approach with other classical and advanced techniques to show the superiority of H2MWSTNet.

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TABLE VI
LAND COVER CATEGORIES OF THE HT DATASET AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CATEGORY

Class No.	Class name	Train	Val	Test
1	Healthy grass	63	62	1126
2	Stressed grass	63	62	1129
3	Synthetic grass	35	35	627
4	Trees	62	62	1120
5	Soil	62	62	1118
6	Water	16	17	292
7	Residential	63	64	1141
8	Commercial	62	62	1120
9	Road	63	62	1127
10	Highway	61	62	1104
11	Railway	62	61	1112
12	Parking Lot 1	61	62	1110
13	Parking Lot 2	24	23	422
14	Tennis Court	21	22	385
15	Running Track	33	33	594
	Total	751	751	13527

TABLE VII
OA (%) FOR DIFFERENT STAGES ON THE FOUR DATASETS

Dataset	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
SA	97.82	97.98	**98.10**	97.55	97.47
PU	98.83	98.91	**99.11**	98.76	98.47
WHU-LK	99.55	99.58	**99.60**	99.55	99.49
HR-L	84.14	84.30	**85.12**	84.48	83.80

TABLE VIII
OA (%) FOR DIFFERENT DEPTHS OF MWSFORMER ENCODER BLOCKS AT EACH STAGE ON THE FOUR DATASETS

Depth	SA	PU	WHU-LK	HR-L
1+1+1	98.04	98.83	99.53	84.67
1+1+2	97.95	98.92	99.54	84.88
1+2+1	97.87	**99.11**	99.55	**85.12**
1+2+2	97.96	98.80	99.51	84.76
2+1+1	**98.10**	98.98	99.57	85.02
2+1+2	98.01	98.90	99.57	84.84
2+2+1	97.93	98.97	**99.60**	84.78
2+2+2	97.82	99.03	99.54	84.56

TABLE IX
OA (%) FOR DIFFERENT HEADS OF MWSFORMER ENCODER BLOCKS AT EACH STAGE ON THE FOUR DATASETS

Head	SA	PU	WHU-LK	HR-L
4+2+1	97.92	98.93	99.56	84.46
8+4+2	**98.10**	**99.11**	**99.60**	**85.12**
16+8+4	97.97	98.93	99.52	84.69
32+16+8	97.66	98.84	99.51	84.38

TABLE X
OA (%) OF MAJOR COMPONENTS ON FOUR DATASETS

ResNet	DBSSC	MHSA	GMSA	SA	PU	WHU-LK	HR-L
✗	✗	✗	✗	96.85	97.77	99.25	82.89
✓	✗	✗	✗	92.64	97.46	98.53	80.14
✗	✓	✗	✗	97.96	98.97	99.54	84.61
✗	✗	✓	✗	94.57	95.24	98.78	77.84
✗	✗	✗	✓	97.85	98.70	99.48	83.02
✓	✗	✓	✗	91.99	97.36	98.60	79.98
✓	✗	✗	✓	92.92	97.59	98.80	80.55
✗	✓	✓	✗	96.91	97.70	99.17	82.28
✗	✓	✗	✓	**98.10**	**99.11**	**99.60**	**85.12**

TABLE XI
OA (%) OF DIFFERENT COMPONENTS IN DBSSC ON FOUR DATASETS

Structure	SA	PU	WHU-LK	HR-L
Without SpeC	97.68	98.80	99.47	84.38
Without SpaC	97.89	98.91	99.48	83.85
DBSSC	**97.96**	**98.97**	**99.54**	**84.61**

A. Data Description

To further verify the classification performance of H2MWSTNet, we selected six classic hyperspectral datasets for our experiments, including Salinas, Pavia University, WHU-LongKou, HyRANK-Loukia, Indian Pines, and Houston 2013.

1) Salinas (SA) Dataset: In 1998, the dataset was acquired in the USA using an AVIRIS sensor. The original SA dataset consists of 224 spectral bands, ranging from 400 to 2500 nm. After removing 20 water absorption bands, 204 spectral bands are retained for research and evaluation. The dataset consists of 512 × 217 pixels and includes 16 vegetation categories, for a total of 54 129 samples.

2) Pavia University (PU) Dataset: In 2001, the dataset was acquired over Pavia in Northern Italy during a flight campaign using the ROSIS sensor. The dataset comprises 115 spectral bands with wavelengths ranging from 430 to 860 nm. After discarding 12 noise bands, 103 spectral bands are retained for research purposes. The image is composed of 610 × 340 pixels, with a total of 42 776 labeled samples across nine land cover types.

3) WHU-LongKou (WHU-LK) Dataset: In 2018, the dataset was acquired in Hubei Province, China, using a DJI M600 Pro platform. The dataset comprises 270 spectral bands, spanning a wavelength range from 400 to 1000 nm. The image is composed of 550 × 400 pixels, covering nine different land cover categories with a total of 204 542 labeled samples.

4) HyRANK-Loukia (HR-L) Dataset: The dataset was acquired from an EO-1 Hyperion sensor. The dataset consists of 176 spectral bands and 249 × 945 pixels, encompassing 14 different land cover categories with a total of 13 503 manually labeled samples.

5) Indian Pines (IP) Dataset: In 1992, the dataset was acquired by the AVIRIS sensor. The original IP dataset consisted of 220 spectral bands, ranging from 400 to 2500 nm. Two hundred spectral bands were retained for HSI classification after the removal of bad and absorbing bands. The dataset consists of 145 × 145 pixels and includes 16 vegetation classes with a total of 10 249 samples.

6) Houston 2013 (HT) Dataset: In 2012, the dataset was acquired in the USA using an ITRES CASI-1500 sensor. The IP dataset consists of 144 bands ranging from 380 to 1050 nm. The dataset consists of 349 × 1905 pixels and includes 15 different land cover classes with a total of 15 029 samples.

In the experiments, we used different proportions to divide the labeled samples in the datasets. Specifically, for the SA and PU datasets, we chose 1% of the labeled samples for training, another 1% for validation, and the remaining 98% for testing at random. For the WHU-LK dataset, 0.5% of the labeled

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TABLE XII
OA (%) OF DIFFERENT ATTENTIONS IN MWSFORMER ON FOUR DATASETS

Attention	SA	PU	WHU-LK	HR-L
OSA	95.78	96.62	99.22	81.04
MWSA	97.11	98.47	99.36	81.98
GFE+OSA	97.40	98.42	99.42	82.27
GFE+MWSA	**97.85**	**98.70**	**99.48**	**83.02**

TABLE XIII
OA (%) OF DIFFERENT COMPONENTS IN MWSFORMER ON FOUR DATASETS

Configuration	SA	PU	WHU-LK	HR-L
Without LFE	97.35	98.22	99.22	81.47
Without LEPE	97.47	98.47	99.43	82.68
Without LFE & LEPE	97.10	98.15	99.18	81.17
LFE + LEPE	**97.85**	**98.70**	**99.48**	**83.02**

TABLE XIV
OA (%) WITH AND WITHOUT HIERARCHICAL STRUCTURE ON FOUR DATASETS

	Metric	SA	PU	WJU-LK	HR-L
Nonhierarchical	OA(%)	97.91	98.96	99.52	84.57
	Params(M)	7.69	7.23	8.62	7.55
Hierarchical	OA(%)	**98.10**	**99.11**	**99.60**	**85.12**
	Params(M)	**3.99**	**3.04**	**4.43**	**3.36**

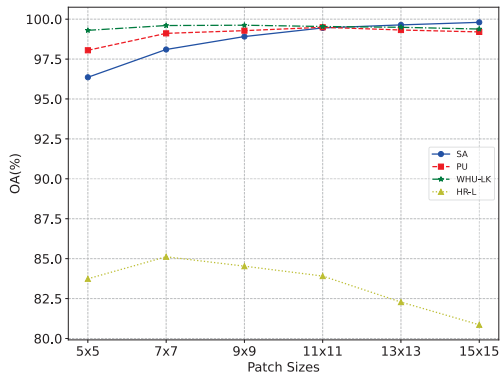


Fig. 4. OA (%) for different patch sizes on the four datasets.

samples were chosen for training, another 0.5% for validation, and the remaining 99% for testing. For the HR-L dataset, 3% of the labeled samples were chosen for training, another 3% for validation, and the remaining 94% for testing. As for the IP and HT dataset, 5% of the labeled samples were assigned to training, another 5% to validation, and the rest, 90%, to testing. Detailed information about the labeled samples allocated for training, validation, and testing across each category in these four datasets is provided in Tables I–VI.

B. Experimental Setup

1) Evaluation Metrics: To comprehensively evaluate the superiority of our method in terms of classification performance, we employed three commonly used evaluation metrics: overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ). The higher the value of each metric, the better the classification results of the model. In addition, to reduce the effect of experimental randomization as well as for fair comparisons, we randomly divided the labeled samples five times and calculated the mean and standard deviation.

2) Comparative Methods: To assess the effectiveness of the proposed H2MWSTNet, we conducted comparative experiments with methods based on CNN, methods based on Transformer, and hybrid methods that combine CNN and Transformer. This includes HybridSN [14], SSRN [13], DBDA [22], residual spectral–spatial attention network (RSSAN) [15], ViT [27], SpectralFormer [30], SSFTT [32], GAHT [36], SSWinT [41], MVAHN [34], morphFormer [38], and MHCFormer [56].

1) For HybridSN, there are three 3-D convolution blocks, one 2-D convolution block, and three fully connected layers.
2) For SSRN, it utilizes the original 3-D HSI cube as input data and employs two residual blocks to extract spectral–spatial features.
3) For DBDA, two branches are designed to extract spatial–spectral features. In the spectral branch, there are five 3-D convolutional blocks and a channel attention module. In the spatial branch, there are four 3-D convolutional blocks and one spatial attention module.
4) For RSSAN, the original 3-D cube is used as the input data, and spectral and spatial attention modules are designed to focus on spectral and spatial features, respectively.
5) For ViT, it utilizes a pure Transformer architecture to perform the image classification task. Set the number of Transformer encoders to 12, with token embeddings of length 64, and the number of heads in the multihead attention mechanism to 8.
6) For SpectralFormer, the input token sequence passes through five Transformer encoders, each with four attention heads, and the dimension of token embeddings is 64.
7) For SSFTT, a $3 \times 3 \times 3$ 3-D convolution and a 3×3 2-D convolution are used to extract shallow spatial–spectral features, respectively.
8) For GAHT, it is divided into three layers, with channels decreasing progressively at each layer. Each layer features a grouped convolution module and several Transformer modules.
9) For SSWinT, it applies Swin Transformer to HSI classification using the sliding window attention mechanism and reduces the network structure from four stages to one, thus reducing the complexity and greatly improving the accuracy of the classification results.
10) For MVAHN, a spectral–spacial evolution module is initially used to extract local information. Subsequently, these features are embedded into the ViT and graph convolution modules through convolutional operations to extract rich features and explore the pixelwise similarities in HSI.
11) For morphFormer, it utilizes 3-D convolution and heterogeneous convolution to extract robust features, where the 3-D convolution and HetConv consist of grouped convolution and pointwise convolution in parallel. Subsequently, these features are fed into a ViT encoder,

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TABLE XV

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE SA DATASET

Class No.	CNN-based Method		CNN-based Method with Attention		Transformer-based Method		Hybrid CNN-Transformer Method						Our Method
	HybridSN [14]	SSRN [13]	DBDA [22]	RSSAN [15]	ViT [27]	SpectralFormer [30]	SSFTT [32]	GAHT [36]	SSWinT [41]	MVAHN [34]	morphFormer [38]	MHCFormer [56]	**H2MWSTNet**
1	98.63 ± 1.32	99.67 ± 0.66	99.99 ± 0.02	98.76 ± 1.05	94.81 ± 2.61	95.88 ± 1.95	99.78 ± 0.44	99.98 ± 0.02	88.72 ± 3.88	99.86 ± 0.28	**100.00 ± 0.00**	76.39 ± 27.60	99.95 ± 0.10
2	99.97 ± 0.05	**100.00 ± 0.00**	**100.00 ± 0.00**	99.80 ± 0.24	96.40 ± 2.99	99.47 ± 0.34	99.86 ± 0.15	99.99 ± 0.01	91.60 ± 4.19	**100.00 ± 0.00**	99.95 ± 0.06	**100.00 ± 0.00**	99.99 ± 0.01
3	96.89 ± 2.03	99.80 ± 0.30	99.50 ± 0.53	97.22 ± 0.98	89.14 ± 2.22	96.59 ± 1.48	98.91 ± 0.37	99.14 ± 0.58	95.72 ± 2.26	96.95 ± 4.33	96.06 ± 4.03	78.62 ± 8.15	**99.83 ± 0.27**
4	97.03 ± 3.32	99.61 ± 0.25	99.14 ± 0.48	96.53 ± 3.02	91.78 ± 7.38	94.39 ± 3.08	99.33 ± 0.66	98.85 ± 1.70	95.84 ± 4.57	98.41 ± 0.86	99.58 ± 0.27	**99.66 ± 0.19**	99.38 ± 0.66
5	97.10 ± 1.24	97.70 ± 1.99	98.63 ± 1.52	95.26 ± 2.97	95.50 ± 3.65	94.65 ± 4.25	96.29 ± 1.71	98.96 ± 1.10	96.63 ± 2.63	**99.61 ± 0.55**	98.74 ± 0.96	98.57 ± 0.88	99.15 ± 1.07
6	**100.00 ± 0.00**	99.97 ± 0.05	99.94 ± 0.12	99.20 ± 1.35	99.42 ± 0.28	99.62 ± 0.37	99.88 ± 0.16	**100.00 ± 0.00**	99.81 ± 0.11	99.92 ± 0.15	99.94 ± 0.11	99.99 ± 0.02	**100.00 ± 0.00**
7	99.20 ± 1.26	99.97 ± 0.03	99.89 ± 0.13	98.95 ± 0.76	94.77 ± 3.44	98.28 ± 0.83	99.61 ± 0.22	99.90 ± 0.09	99.13 ± 1.10	99.92 ± 0.12	99.93 ± 0.10	99.96 ± 0.03	**99.98 ± 0.02**
8	91.96 ± 2.42	94.01 ± 1.36	94.01 ± 0.76	89.09 ± 1.71	84.87 ± 1.48	87.26 ± 2.12	89.67 ± 1.63	92.41 ± 2.60	**97.67 ± 0.80**	94.97 ± 2.68	95.24 ± 2.27	90.63 ± 0.87	95.41 ± 1.41
9	99.89 ± 0.18	99.76 ± 0.33	99.80 ± 0.24	98.28 ± 0.75	94.99 ± 2.60	98.37 ± 1.37	99.38 ± 0.57	**99.98 ± 0.03**	98.11 ± 1.51	99.94 ± 0.13	99.45 ± 0.81	99.95 ± 0.10	99.66 ± 0.67
10	96.64 ± 1.49	96.25 ± 3.08	97.92 ± 1.67	93.57 ± 2.07	85.88 ± 6.51	91.82 ± 2.56	93.89 ± 2.21	97.88 ± 1.52	96.79 ± 1.00	98.15 ± 1.09	97.85 ± 2.04	97.54 ± 1.81	**98.41 ± 1.06**
11	99.03 ± 1.49	99.46 ± 0.31	**99.72 ± 0.17**	97.32 ± 1.63	85.00 ± 7.69	89.44 ± 4.67	96.33 ± 2.94	98.60 ± 1.18	98.26 ± 0.88	99.14 ± 1.32	99.36 ± 0.32	55.04 ± 30.04	99.40 ± 0.34
12	99.78 ± 0.27	99.98 ± 0.04	99.98 ± 0.03	98.63 ± 2.56	97.25 ± 2.00	99.13 ± 0.64	99.70 ± 0.40	99.89 ± 0.12	97.99 ± 0.99	99.98 ± 0.03	99.97 ± 0.04	99.98 ± 0.04	**100.00 ± 0.00**
13	99.19 ± 1.10	99.26 ± 0.91	98.73 ± 1.60	97.75 ± 2.32	96.38 ± 2.53	95.13 ± 5.76	97.07 ± 2.33	97.84 ± 2.54	96.18 ± 3.04	**99.74 ± 0.35**	98.86 ± 1.56	78.76 ± 38.82	98.82 ± 1.70
14	97.59 ± 2.18	98.69 ± 0.50	99.36 ± 0.12	95.25 ± 2.34	90.36 ± 6.28	96.65 ± 1.86	94.28 ± 3.29	98.90 ± 0.51	94.11 ± 3.69	**99.63 ± 0.13**	97.57 ± 1.80	97.89 ± 1.46	97.79 ± 3.07
15	91.78 ± 3.27	92.47 ± 2.66	92.40 ± 3.38	85.00 ± 4.29	79.39 ± 2.25	83.02 ± 4.71	80.61 ± 3.76	91.28 ± 2.60	**97.13 ± 1.41**	95.83 ± 2.11	96.07 ± 1.45	80.68 ± 7.23	95.33 ± 2.07
16	95.42 ± 6.68	97.73 ± 1.92	97.73 ± 2.44	95.73 ± 2.90	81.22 ± 8.54	91.51 ± 5.64	95.51 ± 1.49	98.14 ± 1.78	94.18 ± 6.54	98.48 ± 2.76	97.42 ± 4.72	97.03 ± 1.30	**98.76 ± 0.66**
OA(%)	96.33 ± 0.35	97.22 ± 0.33	97.35 ± 0.44	94.14 ± 0.80	89.72 ± 1.19	92.85 ± 0.71	94.11 ± 0.35	96.85 ± 0.64	96.74 ± 0.59	98.01 ± 0.53	97.89 ± 0.62	92.17 ± 1.90	**98.10 ± 0.24**
AA(%)	97.51 ± 0.57	98.40 ± 0.28	98.55 ± 0.21	96.02 ± 0.70	91.07 ± 1.43	94.45 ± 0.37	96.26 ± 0.29	98.23 ± 0.35	96.12 ± 0.72	98.78 ± 0.39	98.50 ± 0.46	90.67 ± 3.85	**98.87 ± 0.16**
$\kappa \times 100$	95.91 ± 0.39	96.90 ± 0.37	97.05 ± 0.49	93.47 ± 0.89	88.55 ± 1.33	92.04 ± 0.79	93.44 ± 0.39	96.49 ± 0.72	96.37 ± 0.65	97.78 ± 0.59	97.65 ± 0.70	91.27 ± 2.13	**97.88 ± 0.26**

TABLE XVI

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE PU DATASET

Class No.	CNN-based Method		CNN-based Method with Attention		Transformer-based Method		Hybrid CNN-Transformer Method						Our Method
	HybridSN [14]	SSRN [13]	DBDA [22]	RSSAN [15]	ViT [27]	SpectralFormer [30]	SSFTT [32]	GAHT [36]	SSWinT [41]	MVAHN [34]	morphFormer [38]	MHCFormer [56]	**H2MWSTNet**
1	97.48 ± 0.95	**98.67 ± 0.71**	98.36 ± 1.35	96.67 ± 1.70	82.70 ± 2.19	88.67 ± 2.52	96.86 ± 0.49	97.13 ± 1.22	92.48 ± 2.81	98.42 ± 0.97	98.48 ± 0.31	98.13 ± 0.74	98.47 ± 0.51
2	98.35 ± 0.94	99.43 ± 0.28	99.56 ± 0.37	99.46 ± 0.32	94.39 ± 1.40	96.65 ± 1.76	99.50 ± 0.27	99.51 ± 0.12	96.51 ± 2.70	**99.91 ± 0.12**	99.73 ± 0.18	99.85 ± 0.10	99.90 ± 0.06
3	85.86 ± 7.13	91.28 ± 3.95	95.80 ± 1.21	82.64 ± 4.13	47.62 ± 3.68	75.62 ± 8.45	88.98 ± 2.60	88.32 ± 3.28	78.84 ± 8.61	94.51 ± 3.65	94.39 ± 1.75	56.79 ± 8.43	**95.82 ± 1.42**
4	97.92 ± 1.19	96.85 ± 2.51	96.97 ± 1.36	94.67 ± 4.23	85.74 ± 6.40	91.92 ± 3.50	97.29 ± 0.42	97.61 ± 0.42	82.37 ± 2.69	94.65 ± 2.21	96.51 ± 0.49	97.39 ± 1.11	**98.23 ± 0.68**
5	99.97 ± 0.06	**100.00 ± 0.00**	99.90 ± 0.14	99.03 ± 0.85	94.56 ± 2.47	**100.00 ± 0.00**	99.96 ± 0.09	**100.00 ± 0.00**	93.70 ± 5.80	99.61 ± 0.49	99.84 ± 0.17	99.99 ± 0.03	**100.00 ± 0.00**
6	91.94 ± 4.78	99.31 ± 0.59	99.07 ± 0.53	96.25 ± 1.69	58.76 ± 6.74	83.28 ± 5.13	98.37 ± 0.50	98.19 ± 1.59	95.62 ± 2.68	99.65 ± 0.20	**99.87 ± 0.10**	98.06 ± 1.53	99.20 ± 0.58
7	81.20 ± 6.97	98.18 ± 1.83	98.20 ± 1.77	87.67 ± 5.32	35.79 ± 11.79	62.35 ± 6.13	92.18 ± 2.55	93.07 ± 7.43	83.70 ± 11.54	**99.68 ± 0.42**	95.77 ± 4.66	70.21 ± 6.91	98.53 ± 2.30
8	95.08 ± 2.96	95.19 ± 2.49	97.77 ± 1.15	91.28 ± 2.73	71.13 ± 6.74	78.98 ± 5.52	93.36 ± 1.75	95.35 ± 1.80	83.98 ± 12.09	97.54 ± 1.27	95.50 ± 1.35	98.26 ± 0.69	**98.42 ± 0.64**
9	99.62 ± 0.39	**99.94 ± 0.05**	99.60 ± 0.34	98.50 ± 0.82	85.77 ± 9.71	96.96 ± 0.99	99.51 ± 0.36	99.77 ± 0.12	46.19 ± 16.15	96.18 ± 3.38	96.37 ± 1.29	97.55 ± 2.01	99.87 ± 0.10
OA(%)	96.08 ± 0.93	98.34 ± 0.22	98.76 ± 0.31	96.37 ± 0.73	81.46 ± 1.34	89.99 ± 0.81	97.54 ± 0.29	97.77 ± 0.13	91.22 ± 3.55	98.70 ± 0.19	98.50 ± 0.28	95.98 ± 0.48	**99.11 ± 0.14**
AA(%)	94.16 ± 1.60	97.65 ± 0.62	98.36 ± 0.24	94.02 ± 1.30	72.94 ± 3.16	86.05 ± 1.16	96.22 ± 0.59	96.55 ± 0.66	83.71 ± 5.76	97.79 ± 0.46	97.38 ± 0.77	90.69 ± 0.80	**98.71 ± 0.29**
$\kappa \times 100$	94.80 ± 1.24	97.80 ± 0.30	98.36 ± 0.41	95.18 ± 0.97	75.23 ± 1.90	86.70 ± 1.07	96.74 ± 0.38	97.04 ± 0.17	88.38 ± 4.63	98.28 ± 0.25	98.01 ± 0.37	94.65 ± 0.65	**98.82 ± 0.19**

TABLE XVII

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE WHU-LK DATASET

Class No.	CNN-based Method		CNN-based Method with Attention		Transformer-based Method		Hybrid CNN-Transformer Method						Our Method
	HybridSN [14]	SSRN [13]	DBDA [22]	RSSAN [15]	ViT [27]	SpectralFormer [30]	SSFTT [32]	GAHT [36]	SSWinT [41]	MVAHN [34]	morphFormer [38]	MHCFormer [56]	**H2MWSTNet**
1	99.79 ± 0.17	99.88 ± 0.07	99.87 ± 0.07	99.70 ± 0.13	97.43 ± 0.72	99.59 ± 0.14	99.78 ± 0.10	99.82 ± 0.15	99.80 ± 0.09	99.93 ± 0.03	99.88 ± 0.07	99.93 ± 0.04	**99.96 ± 0.02**
2	95.60 ± 2.88	98.82 ± 1.04	98.11 ± 0.45	97.14 ± 0.89	83.90 ± 3.10	92.09 ± 2.36	96.78 ± 2.17	98.24 ± 1.05	98.13 ± 1.09	99.09 ± 0.77	98.97 ± 0.59	99.31 ± 0.25	**99.58 ± 0.22**
3	94.99 ± 3.19	98.69 ± 0.86	98.15 ± 1.96	94.56 ± 2.00	32.99 ± 13.93	95.06 ± 1.46	94.79 ± 1.65	96.77 ± 1.51	93.05 ± 9.76	96.23 ± 2.75	98.01 ± 1.22	95.22 ± 2.05	**99.04 ± 0.52**
4	98.61 ± 0.80	99.49 ± 0.14	99.70 ± 0.09	99.06 ± 0.24	96.46 ± 1.23	98.37 ± 0.38	99.30 ± 0.24	99.57 ± 0.27	98.96 ± 0.36	**99.75 ± 0.10**	99.48 ± 0.29	99.71 ± 0.13	99.66 ± 0.13
5	88.34 ± 6.95	**96.76 ± 2.15**	95.46 ± 1.16	90.76 ± 4.09	45.49 ± 23.59	83.87 ± 2.61	90.58 ± 2.54	91.93 ± 2.29	91.56 ± 2.67	95.80 ± 1.70	94.94 ± 2.14	92.41 ± 1.86	95.94 ± 1.64
6	98.63 ± 1.10	99.72 ± 0.19	**99.92 ± 0.07**	98.84 ± 1.12	90.05 ± 1.60	98.06 ± 0.81	99.60 ± 0.31	99.79 ± 0.29	99.40 ± 0.31	99.54 ± 0.23	99.13 ± 0.68	99.78 ± 0.15	99.88 ± 0.11
7	**99.99 ± 0.01**	99.95 ± 0.02	99.86 ± 0.07	99.98 ± 0.01	99.88 ± 0.09	99.93 ± 0.06	99.98 ± 0.02	99.96 ± 0.03	99.80 ± 0.18	99.92 ± 0.04	99.92 ± 0.07	99.97 ± 0.01	99.98 ± 0.01
8	95.13 ± 1.97	**97.90 ± 0.93**	97.33 ± 1.12	93.90 ± 2.97	81.81 ± 5.30	88.94 ± 3.55	95.65 ± 2.48	97.33 ± 1.12	93.64 ± 2.01	93.75 ± 2.41	95.19 ± 2.57	97.33 ± 1.21	97.89 ± 1.14
9	93.43 ± 3.02	94.42 ± 1.14	93.74 ± 2.56	92.11 ± 2.29	87.60 ± 1.04	82.32 ± 4.26	92.69 ± 2.88	92.77 ± 2.07	88.76 ± 3.21	91.27 ± 3.43	90.55 ± 4.24	94.81 ± 1.83	**96.60 ± 1.89**
OA(%)	98.62 ± 0.51	99.44 ± 0.11	99.38 ± 0.07	98.78 ± 0.11	94.15 ± 0.21	97.73 ± 0.22	98.98 ± 0.17	99.25 ± 0.19	98.69 ± 0.22	99.24 ± 0.14	99.16 ± 0.14	99.40 ± 0.08	**99.60 ± 0.09**
AA(%)	96.06 ± 1.10	98.41 ± 0.31	98.01 ± 0.28	96.23 ± 0.58	79.51 ± 3.48	93.14 ± 0.57	96.57 ± 0.51	97.35 ± 0.60	95.90 ± 1.10	97.25 ± 0.64	97.34 ± 0.41	97.61 ± 0.28	**98.73 ± 0.29**
$\kappa \times 100$	98.19 ± 0.67	99.27 ± 0.15	99.19 ± 0.09	98.40 ± 0.14	92.28 ± 0.30	97.01 ± 0.29	98.65 ± 0.22	99.01 ± 0.26	98.27 ± 0.29	99.00 ± 0.18	98.89 ± 0.18	99.21 ± 0.11	**99.47 ± 0.12**

TABLE XVIII

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HR-L DATASET

Class No.	CNN-based Method		CNN-based Method with Attention		Transformer-based Method		Hybrid CNN-Transformer Method						Our Method
	HybridSN [14]	SSRN [13]	DBDA [22]	RSSAN [15]	ViT [27]	SpectralFormer [30]	SSFTT [32]	GAHT [36]	SSWinT [41]	MVAHN [34]	morphFormer [38]	MHCFormer [56]	**H2MWSTNet**
1	38.33 ± 9.49	59.65 ± 9.41	61.60 ± 6.69	56.94 ± 9.45	7.92 ± 7.19	47.99 ± 8.51	60.83 ± 10.10	64.44 ± 11.15	37.64 ± 10.07	38.26 ± 7.31	**65.49 ± 6.72**	14.44 ± 14.76	64.17 ± 13.82
2	72.54 ± 38.77	96.42 ± 3.35	**100.00 ± 0.00**	78.51 ± 10.87	31.34 ± 31.70	95.82 ± 5.12	79.70 ± 14.94	92.24 ± 8.41	15.22 ± 22.07	86.57 ± 8.70	92.24 ± 6.43	3.58 ± 7.16	94.33 ± 8.14
3	75.79 ± 11.41	83.80 ± 4.45	87.68 ± 4.10	74.83 ± 10.20	53.32 ± 8.07	65.65 ± 7.87	81.73 ± 4.94	**88.56 ± 3.67**	50.00 ± 10.05	76.61 ± 4.04	82.21 ± 4.96	81.66 ± 8.64	85.39 ± 2.61
4	16.46 ± 10.28	43.54 ± 8.64	46.84 ± 16.85	32.41 ± 9.01	0.51 ± 1.01	14.94 ± 7.86	47.85 ± 6.12	39.75 ± 11.25	26.84 ± 11.22	1.01 ± 2.03	34.94 ± 7.10	0.25 ± 0.51	**54.43 ± 7.51**
5	83.04 ± 2.82	89.01 ± 2.22	90.82 ± 0.97	86.40 ± 4.21	35.40 ± 12.58	74.10 ± 4.52	88.69 ± 1.93	89.55 ± 3.49	63.94 ± 16.23	91.85 ± 4.72	87.57 ± 1.79	**95.27 ± 1.06**	92.43 ± 1.40
6	42.15 ± 4.71	43.23 ± 11.80	43.32 ± 12.04	45.83 ± 8.23	25.29 ± 7.87	25.74 ± 6.28	57.49 ± 16.66	47.53 ± 12.88	14.53 ± 10.22	9.60 ± 7.43	41.26 ± 9.95	4.75 ± 3.75	**61.26 ± 10.69**
7	53.92 ± 9.80	62.68 ± 4.13	**64.92 ± 5.14**	53.00 ± 4.22	47.24 ± 8.93	45.32 ± 2.98	59.76 ± 5.48	62.28 ± 10.94	40.60 ± 6.11	40.52 ± 5.31	61.28 ± 8.86	44.40 ± 10.99	64.44 ± 5.60
8	67.87 ± 10.10	75.95 ± 7.45	**79.85 ± 5.93**	67.13 ± 5.54	49.33 ± 6.27	60.19 ± 3.38	73.88 ± 5.43	70.09 ± 9.84	62.29 ± 11.50	61.38 ± 10.94	76.10 ± 5.55	71.19 ± 3.79	78.13 ± 4.30
9	83.94 ± 3.15	85.26 ± 5.21	83.70 ± 2.92	83.53 ± 2.24	73.01 ± 5.74	72.40 ± 6.91	82.60 ± 2.78	83.45 ± 2.25	78.64 ± 1.36	79.05 ± 3.45	81.50 ± 2.55	84.25 ± 1.54	**85.47 ± 2.09**
10	80.09 ± 3.62	79.81 ± 6.14	78.00 ± 5.19	81.39 ± 1.76	72.17 ± 6.29	72.84 ± 5.84	81.51 ± 3.47	81.06 ± 2.02	70.37 ± 6.99	73.90 ± 2.22	77.97 ± 4.26	82.66 ± 2.59	**83.01 ± 1.31**
11	56.24 ± 11.32	75.45 ± 6.75	75.20 ± 6.31	62.48 ± 13.01	45.64 ± 13.21	62.03 ± 5.28	69.60 ± 6.62	67.77 ± 3.60	47.57 ± 19.59	52.13 ± 1.90	**75.59 ± 3.38**	17.28 ± 11.43	68.86 ± 10.08
12	89.12 ± 5.74	92.03 ± 3.94	92.44 ± 5.24	87.23 ± 5.81	75.98 ± 11.56	89.90 ± 6.08	92.61 ± 2.58	93.47 ± 1.77	82.63 ± 11.48	80.62 ± 7.41	92.03 ± 4.95	92.65 ± 2.39	**94.25 ± 3.04**
13	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
14	99.78 ± 0.44	**100.00 ± 0.00**	**100.00 ± 0.00**	99.38 ± 1.03	99.11 ± 1.25	**100.00 ± 0.00**	**100.00 ± 0.00**	99.87 ± 0.27	98.89 ± 1.27	99.16 ± 1.69	99.42 ± 0.79	**100.00 ± 0.00**	99.82 ± 0.35
OA(%)	79.76 ± 1.42	83.42 ± 0.79	83.43 ± 1.11	80.82 ± 1.11	65.44 ± 1.40	73.00 ± 0.66	82.69 ± 0.54	82.89 ± 0.51	71.09 ± 3.70	76.03 ± 0.64	81.73 ± 0.58	79.20 ± 0.51	**85.12 ± 0.10**
AA(%)	68.52 ± 3.69	77.63 ± 1.66	78.88 ± 2.46	72.07 ± 2.11	51.16 ± 5.12	66.21 ± 0.08	76.88 ± 1.15	77.15 ± 1.12	56.37 ± 3.64	63.62 ± 1.53	76.26 ± 1.17	56.60 ± 2.26	**80.43 ± 0.90**
$\kappa \times 100$	75.74 ± 1.77	80.22 ± 0.89	80.23 ± 1.36	77.06 ± 1.38	58.22 ± 1.81	67.85 ± 0.65	79.36 ± 0.66	79.58 ± 0.63	65.17 ± 4.54	71.14 ± 0.83	78.22 ± 0.75	74.95 ± 0.68	**82.29 ± 0.10**

and classification results are obtained through a classifier head.

12) For MHCFormer, it is structured into three layers. Initially, 3-D convolution is employed to enhance both spatial and spectral features of HSI. HetConv is utilized to effectively integrate spatial–frequency features. Subsequently, these features are fed into a FourierFormer encoder block, with FourierMixer at its core. Finally, classification is performed in the fully connected layer.

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TABLE XIX
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE IP DATASET

Class No.	CNN-based Method		CNN-based Method with Attention		Transformer-based Method		Hybrid CNN-Transformer Method						Our Method
	HybridSN [14]	SSRN [13]	DBDA [22]	RSSAN [15]	ViT [27]	SpectralFormer [30]	SSFTT [32]	GAHT [36]	SSWinT [41]	MVAHN [34]	morphFormer [38]	MHCFormer [56]	**H2MWSTNet**
1	66.96 ± 12.25	**92.61 ± 4.48**	87.39 ± 7.70	58.70 ± 17.93	40.87 ± 15.46	27.83 ± 4.64	68.70 ± 21.55	73.48 ± 18.97	51.74 ± 25.55	87.39 ± 14.38	78.26 ± 8.59	1.30 ± 1.74	92.61 ± 6.09
2	89.41 ± 2.88	96.79 ± 1.61	95.49 ± 1.96	90.15 ± 2.37	66.88 ± 3.10	75.22 ± 1.18	91.88 ± 2.66	94.78 ± 1.83	79.52 ± 3.25	95.74 ± 2.11	95.84 ± 2.54	95.03 ± 1.69	**97.41 ± 1.45**
3	89.66 ± 5.16	94.82 ± 3.28	96.27 ± 2.66	92.58 ± 1.36	56.29 ± 7.62	74.39 ± 2.71	89.88 ± 2.79	92.46 ± 3.84	78.80 ± 8.79	96.14 ± 1.65	95.47 ± 3.00	95.33 ± 2.75	**97.04 ± 2.71**
4	85.65 ± 3.41	93.59 ± 8.00	**95.02 ± 3.11**	71.05 ± 8.21	47.85 ± 7.94	61.18 ± 6.18	82.78 ± 5.09	89.79 ± 5.38	73.67 ± 19.78	94.68 ± 1.72	92.32 ± 5.04	86.24 ± 5.76	94.60 ± 3.72
5	91.64 ± 2.42	91.72 ± 2.77	91.64 ± 3.31	88.86 ± 2.80	61.61 ± 9.55	79.13 ± 4.30	91.06 ± 3.85	92.92 ± 1.49	71.72 ± 9.12	89.86 ± 3.90	92.05 ± 2.64	93.08 ± 3.07	**93.17 ± 3.51**
6	98.85 ± 0.56	98.96 ± 1.10	99.01 ± 1.29	98.55 ± 1.01	83.81 ± 5.79	94.71 ± 2.02	99.42 ± 0.29	**99.81 ± 0.21**	77.26 ± 8.68	98.90 ± 0.93	99.26 ± 0.66	99.75 ± 0.33	99.78 ± 0.31
7	87.86 ± 13.48	97.14 ± 5.71	95.71 ± 4.16	75.71 ± 28.16	37.86 ± 18.01	57.86 ± 24.16	95.71 ± 4.16	**99.29 ± 1.43**	60.71 ± 19.03	74.29 ± 15.22	79.29 ± 16.81	4.29 ± 5.71	94.29 ± 9.74
8	99.33 ± 0.16	99.67 ± 0.43	99.96 ± 0.08	96.82 ± 1.93	92.93 ± 5.12	95.31 ± 2.69	98.33 ± 1.50	99.79 ± 0.26	95.65 ± 3.52	**100.00 ± 0.00**	99.71 ± 0.36	**100.00 ± 0.00**	99.92 ± 0.17
9	72.00 ± 25.02	77.00 ± 16.00	63.00 ± 20.40	46.00 ± 12.81	20.00 ± 4.47	40.00 ± 10.00	62.00 ± 20.88	76.00 ± 13.19	49.00 ± 24.37	9.00 ± 9.17	47.00 ± 13.27	3.00 ± 6.00	**91.00 ± 7.35**
10	89.63 ± 3.28	93.29 ± 1.78	93.27 ± 5.05	91.05 ± 3.06	67.47 ± 6.10	78.42 ± 0.87	90.10 ± 2.41	93.68 ± 1.75	70.35 ± 13.11	92.98 ± 2.46	95.93 ± 1.27	94.22 ± 1.76	**96.07 ± 0.91**
11	95.46 ± 1.08	96.48 ± 1.10	97.26 ± 1.09	93.29 ± 2.58	78.92 ± 4.18	85.25 ± 1.23	94.38 ± 0.61	97.50 ± 0.98	91.14 ± 2.89	96.81 ± 0.41	97.87 ± 0.69	97.69 ± 0.32	**98.08 ± 0.79**
12	84.45 ± 4.13	95.92 ± 1.87	93.59 ± 3.01	79.33 ± 2.62	48.80 ± 6.38	52.55 ± 5.67	85.36 ± 4.73	94.00 ± 3.44	74.64 ± 7.30	96.16 ± 2.98	94.17 ± 3.74	93.79 ± 2.09	**96.49 ± 2.47**
13	98.44 ± 1.36	99.61 ± 0.57	98.63 ± 0.84	98.24 ± 1.22	86.44 ± 12.49	97.37 ± 2.90	99.51 ± 0.31	**100.00 ± 0.00**	80.29 ± 7.09	98.54 ± 1.27	99.32 ± 1.37	98.54 ± 1.23	99.61 ± 0.37
14	98.56 ± 1.06	99.57 ± 0.32	99.43 ± 0.42	97.45 ± 1.08	93.94 ± 1.48	94.75 ± 1.48	97.71 ± 1.05	98.29 ± 1.15	93.75 ± 3.83	99.24 ± 0.65	98.96 ± 0.87	**99.72 ± 0.22**	99.68 ± 0.23
15	91.87 ± 3.06	86.06 ± 1.44	90.47 ± 5.99	82.54 ± 8.27	74.09 ± 3.76	73.94 ± 4.85	79.95 ± 6.87	81.87 ± 8.97	89.12 ± 4.53	91.19 ± 4.36	**94.35 ± 2.94**	89.74 ± 1.85	91.40 ± 4.64
16	84.30 ± 18.05	**99.35 ± 0.86**	93.98 ± 7.83	88.17 ± 10.02	43.44 ± 10.64	98.06 ± 2.09	94.84 ± 2.92	95.91 ± 3.93	59.35 ± 32.85	91.61 ± 4.63	90.32 ± 8.25	66.24 ± 12.02	96.13 ± 3.23
OA(%)	92.98 ± 1.19	96.11 ± 0.36	96.17 ± 0.66	91.46 ± 1.05	73.15 ± 2.08	81.41 ± 1.04	92.66 ± 0.74	95.42 ± 0.23	82.90 ± 4.53	95.93 ± 0.37	96.50 ± 0.60	95.15 ± 0.31	**97.42 ± 0.44**
AA(%)	89.00 ± 2.02	94.54 ± 1.45	93.13 ± 1.68	84.28 ± 3.00	62.57 ± 2.40	74.12 ± 1.61	88.85 ± 2.40	92.47 ± 1.46	74.79 ± 8.03	88.28 ± 2.08	90.63 ± 1.14	76.12 ± 0.87	**96.08 ± 0.67**
$\kappa \times 100$	91.98 ± 1.37	95.57 ± 0.41	95.63 ± 0.76	90.25 ± 1.19	69.33 ± 2.33	78.79 ± 1.20	91.62 ± 0.85	94.77 ± 0.26	80.48 ± 5.17	95.36 ± 0.42	96.01 ± 0.69	94.46 ± 0.36	**97.06 ± 0.50**

TABLE XX
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HT DATASET

Class No.	CNN-based Method		CNN-based Method with Attention		Transformer-based Method		Hybrid CNN-Transformer Method						Our Method
	HybridSN [14]	SSRN [13]	DBDA [22]	RSSAN [15]	ViT [27]	SpectralFormer [30]	SSFTT [32]	GAHT [36]	SSWinT [41]	MVAHN [34]	morphFormer [38]	MHCFormer [56]	**H2MWSTNet**
1	97.03 ± 1.70	99.01 ± 0.74	98.10 ± 0.91	97.57 ± 1.66	84.99 ± 4.39	96.42 ± 1.92	97.78 ± 1.97	98.83 ± 0.82	86.60 ± 6.25	97.78 ± 1.64	97.99 ± 1.69	92.84 ± 5.93	**99.68 ± 0.64**
2	99.11 ± 0.82	**99.38 ± 0.32**	98.93 ± 0.57	98.68 ± 0.91	84.55 ± 8.00	97.16 ± 1.00	99.28 ± 0.76	99.23 ± 0.49	82.34 ± 7.37	98.66 ± 0.89	99.20 ± 0.72	99.09 ± 1.70	98.85 ± 1.27
3	99.94 ± 0.11	**100.00 ± 0.00**	99.97 ± 0.06	99.54 ± 0.40	98.02 ± 0.88	99.28 ± 0.57	99.97 ± 0.06	**100.00 ± 0.00**	98.28 ± 1.46	99.51 ± 0.28	99.77 ± 0.27	99.94 ± 0.11	**100.00 ± 0.00**
4	95.61 ± 1.27	99.45 ± 1.01	98.89 ± 1.40	96.27 ± 2.28	81.40 ± 4.04	95.14 ± 2.46	98.59 ± 1.56	99.65 ± 0.19	80.90 ± 4.70	97.68 ± 1.46	98.41 ± 1.00	97.48 ± 2.57	**99.68 ± 0.57**
5	99.34 ± 0.80	**100.00 ± 0.00**	99.90 ± 0.06	99.39 ± 0.37	92.53 ± 2.66	98.21 ± 0.46	99.94 ± 0.13	99.87 ± 0.17	98.84 ± 1.52	**100.00 ± 0.00**	99.57 ± 0.50	99.98 ± 0.03	**100.00 ± 0.00**
6	91.14 ± 4.23	96.62 ± 5.43	96.18 ± 5.26	86.95 ± 4.86	60.49 ± 13.87	85.54 ± 5.17	95.82 ± 5.03	95.63 ± 5.08	80.06 ± 4.54	95.82 ± 5.49	95.45 ± 4.52	88.62 ± 2.68	**97.54 ± 4.92**
7	95.30 ± 1.75	96.40 ± 2.04	95.11 ± 2.53	92.00 ± 3.43	91.26 ± 4.76	85.08 ± 3.06	94.29 ± 0.98	94.45 ± 0.76	88.47 ± 2.58	95.47 ± 1.31	95.58 ± 1.26	94.78 ± 1.72	**96.44 ± 1.06**
8	88.26 ± 4.31	94.41 ± 2.87	95.50 ± 3.49	89.73 ± 2.91	63.67 ± 6.71	86.66 ± 3.45	92.77 ± 2.77	93.33 ± 2.83	76.21 ± 4.76	95.77 ± 2.20	**96.09 ± 1.93**	87.67 ± 4.23	94.53 ± 3.07
9	94.36 ± 0.81	95.88 ± 2.61	95.18 ± 1.80	92.33 ± 2.95	87.24 ± 3.32	83.00 ± 1.33	91.84 ± 3.43	95.88 ± 2.28	84.15 ± 2.06	96.09 ± 0.85	94.20 ± 0.66	94.58 ± 0.99	**97.06 ± 1.74**
10	98.00 ± 1.02	99.33 ± 0.34	98.71 ± 0.87	94.05 ± 1.25	83.98 ± 4.32	89.06 ± 1.72	98.92 ± 0.54	97.85 ± 0.88	88.49 ± 3.20	99.90 ± 0.12	98.65 ± 0.96	92.40 ± 1.81	**99.97 ± 0.07**
11	92.28 ± 2.18	96.36 ± 1.03	97.88 ± 0.90	90.53 ± 2.26	76.53 ± 10.35	80.42 ± 2.69	95.89 ± 0.84	96.84 ± 1.26	87.84 ± 10.18	97.57 ± 1.70	97.23 ± 1.72	94.35 ± 1.80	**98.90 ± 1.29**
12	95.25 ± 4.21	97.36 ± 2.38	97.75 ± 1.23	96.72 ± 1.38	72.47 ± 10.63	91.16 ± 2.59	96.92 ± 1.54	96.59 ± 1.37	85.77 ± 5.53	**98.82 ± 0.51**	97.05 ± 2.99	93.53 ± 2.54	98.72 ± 0.45
13	94.07 ± 2.47	94.54 ± 1.54	92.03 ± 4.66	86.74 ± 7.23	79.49 ± 4.36	55.82 ± 5.35	93.90 ± 1.15	96.63 ± 1.22	75.52 ± 5.15	93.56 ± 1.68	95.14 ± 1.70	95.18 ± 0.29	**96.97 ± 1.72**
14	98.08 ± 1.87	99.95 ± 0.09	99.86 ± 0.28	97.48 ± 1.28	79.86 ± 3.97	87.20 ± 4.13	99.25 ± 0.79	99.86 ± 0.19	94.67 ± 5.30	99.91 ± 0.19	99.49 ± 0.54	99.49 ± 0.58	**100.00 ± 0.00**
15	99.79 ± 0.26	**100.00 ± 0.00**	99.97 ± 0.06	98.76 ± 0.63	93.12 ± 5.16	97.67 ± 2.14	98.45 ± 2.65	**100.00 ± 0.00**	95.88 ± 2.06	99.97 ± 0.06	99.70 ± 0.42	**100.00 ± 0.00**	**100.00 ± 0.00**
OA(%)	95.79 ± 0.94	97.89 ± 0.24	97.66 ± 0.37	94.79 ± 0.32	82.54 ± 2.91	89.72 ± 0.22	96.82 ± 0.59	97.52 ± 0.33	86.76 ± 2.83	97.83 ± 0.35	97.55 ± 0.33	95.18 ± 0.71	**98.51 ± 0.29**
AA(%)	95.84 ± 0.83	97.91 ± 0.35	97.60 ± 0.49	94.45 ± 0.58	81.97 ± 2.77	88.52 ± 0.31	96.91 ± 0.69	97.64 ± 0.46	86.94 ± 2.37	97.77 ± 0.54	97.57 ± 0.53	95.33 ± 0.72	**98.56 ± 0.35**
$\kappa \times 100$	95.45 ± 1.01	97.72 ± 0.26	97.47 ± 0.40	94.36 ± 0.35	81.12 ± 3.15	88.88 ± 0.24	96.56 ± 0.64	97.32 ± 0.35	85.68 ± 3.06	97.65 ± 0.38	97.35 ± 0.36	94.78 ± 0.77	**98.39 ± 0.32**

TABLE XXI
COMPLEXITY ANALYSIS RESULTS OF DIFFERENT METHODS ACROSS FOUR DATASETS

Methods	SA				PU				WHU-LK				HR-L			
	Train(m)	Test(m)	Params(M)	FLOPs(M)	Train(m)	Test(m)	Params(M)	FLOPs(M)	Train(m)	Test(m)	Params(M)	FLOPs(M)	Train(m)	Test(m)	Params(M)	FLOPs(M)
HybridSN [14]	20.45	4.33	8.33	2540	8.29	3.99	6.47	1210	50.27	11.77	9.54	3420	13.22	8.10	7.81	2180
SSRN [13]	2.47	0.33	0.37	98.12	1.35	0.34	0.22	49.20	5.85	0.84	0.47	130.41	1.71	0.62	0.33	84.42
DBDA [22]	3.91	0.76	0.39	108.96	1.94	0.78	0.20	69.47	9.37	1.97	0.51	144.46	2.55	1.40	0.34	93.89
RSSAN [15]	1.92	0.14	0.11	4.79	1.35	0.23	0.07	3.33	3.92	0.29	0.13	5.75	1.39	0.30	0.10	4.39
ViT [27]	10.69	5.49	1.70	655.83	5.35	3.24	1.70	290.69	34.02	18.66	1.70	884.70	12.26	9.00	1.70	556.70
SpectralFormer [30]	10.25	1.14	0.37	36.24	3.38	0.59	0.17	14.35	17.65	1.50	0.56	54.78	5.83	1.91	0.30	29.39
SSFTT [32]	1.68	0.30	0.96	81.77	0.45	0.27	0.49	40.94	2.09	0.64	1.26	108.45	0.57	0.48	0.83	70.45
GAHT [36]	2.02	0.22	0.97	47.63	1.35	0.37	0.93	45.42	4.27	0.50	1.51	74.19	1.46	0.48	0.82	40.00
SSWinT [41]	1.59	0.60	0.34	8.94	0.72	0.54	0.34	8.75	4.32	2.13	0.34	9.06	1.69	0.96	0.34	8.88
MVAHN [34]	0.98	0.41	0.61	86.54	1.21	0.66	0.48	70.90	1.91	0.87	0.69	96.76	1.34	0.84	0.57	82.21
morphFormer [38]	8.69	0.68	0.28	36.58	5.45	1.40	0.23	21.71	17.44	1.46	0.33	48.46	5.53	1.34	0.26	31.54
MHCFormer [56]	2.79	0.43	3.91	322.74	1.80	0.74	2.04	167.76	5.73	0.99	5.12	424.01	2.04	0.89	3.39	279.77
H2MWSTNet	3.04	0.57	3.99	184.29	2.27	1.04	3.04	140.13	3.24	1.35	4.43	202.50	2.30	1.20	3.36	148.85

3) Implementation Details: For a fair comparison, the H2MWSTNet we designed was implemented on the PyTorch deep learning framework. We conducted experiments using the NVIDIA TESLA T4 graphics processing unit (GPU) equipped with 16 GB of memory. We set the batch size to 64, with each patch size set to 7×7, and the training process to 200 epochs. We employed the stochastic gradient descent (SGD) optimizer with a weight decay of 0.0001, a momentum of 0.9, and an initial learning rate of 0.001. For the configuration of the MWSFormer encoder architecture, we set the embedding dimensions for the three stages C_1, C_2, and C_3 to 256, 128, and 64, respectively, with attention heads set to 8, 4, and 2. It is worth noting that due to variations in performance across different datasets within the network, the encoder depth varies for each dataset at different stages. For the SA dataset, the encoder depth is set to 2, 1, and 1; for the PU and HR-L datasets, the encoder depth is 1, 2, and 1; and for the WHU-LK dataset, the encoder depth is 2, 2, and 1.

C. Sensitivity Analysis of Parameters

To comprehensively analyze the proposed H2MWSTNet model, we investigated several hyperparameters within the network. We primarily focused on the different stages, the depth of the MWSFormer encoder, the number of attention heads, and the size of the HSI patches. To analyze the impact of these hyperparameters on the model, we employed the control variable method, i.e., the other parameter settings are kept fixed, as shown in the experimental setup.

1) Analysis of Different Stages: We conducted experiments at different numbers of stages. Specifically, we analyzed the effect of the number of stages on the classification results in

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terms of stages 1, 2, 3, 4, and 5. We keep the other parameters unchanged and use the method of decreasing the channels layer by layer to set the channels of each stage to 256, 128, 64, 32, 16, and 8, respectively, which ensures the fairness and impartiality of the experiments and at the same time reduces the redundancy of information. As shown in Table VII, the classification results for the four datasets at different number of stages are demonstrated. We found that when the number of stages is small, the model may not be sufficient to capture complex features in HSIs, resulting in slightly lower classification performance. When the number of stages is too high, the complexity of the model increases, which may lead to overfitting phenomena and some degradation in classification performance. A moderate number of stages balances model complexity and classification performance for optimal results. Therefore, we set the number of stages to 3.

2) Analysis of MWSFormer Hyperparameters: Our network model is divided into three stages, and the depth of the MWSFormer encoder at each stage is crucial for the classification results of our network. However, considering the finite model capacity, we constrain the depth of the MWSFormer encoder at each stage to either 1 or 2, while keeping other parameters fixed. Thus, there are a total of eight comparison scenarios. As shown in Table VIII, the optimal encoder depths vary slightly for different datasets. For the PU and HR-L datasets, the optimal depths for the three stages are 1, 2, and 1, respectively. For the SA dataset, the optimal depths are 2, 1, and 1, while for the WHU-LK dataset, they are 2, 2, and 1.

The number of attention heads is also crucial in our network classification. Our network follows a hierarchical structure with decreasing channels at each layer. Therefore, we set the embedding dimensions of the MWSFormer encoder at each stage to 256, 128, and 64, respectively. The number of attention heads is implemented through channel splitting. Therefore, our setting for the number of attention heads follows a rule of gradually decreasing at each layer, with each stage having half the number of attention heads as the previous stage. We keep other parameters fixed and set the number of attention heads to 32, 16, 8, 4, 2, and 1, respectively, for each stage, resulting in four different comparison scenarios. As shown in Table IX, the optimal configuration for the number of attention heads across the four datasets is 8, 4, and 2. Therefore, we set the number of attention heads to 8, 4, and 2 accordingly.

3) Analysis of Different Patch Sizes: We further evaluated the impact of patch size on the classification results. We selected the optimal patch size from the options of 5×5, 7×7, 9×9, 11×11, 13×13, and 15×15. As shown in Fig. 4, the OA graphs for different patch sizes on the four datasets are demonstrated. We found that the OA values of the SA dataset increase with increasing spatial size; however, the other datasets do not always show an increase in classification accuracy with increasing spatial size. The OA on the WHU-LK and PU datasets starts to decrease when the spatial size is 9 and 11. The main reason is due to the relatively large differences in the regions of these two datasets. When the spatial size is 7, the OA value on the HR-L dataset decreases sharply, which is due to the fact that the regions of this dataset are small and independent. Finally, for a fair comparison, for all datasets, we set the HSI patch size to 7×7.

D. Ablation Study

1) Main Module Ablation Study: To analyze the contributions of the main components within our proposed H2MWSTNet model, we conducted ablation experiments with different module combinations across four datasets. Our model primarily consists of two components: the DBSSC module and the GMSA module. To ensure fairness, we also added a simple CNN network with similar parameters (e.g., ResNet) and similarly a simple Transformer network with similar parameters (e.g., multiheaded self-attention (MHSA) in the GAHT paper) for the ablation experiments. As shown in Table X, we present nine combinations of these four structures. Overall, the model performs worst when four modules are omitted, regardless of the dataset used. In our proposed model, the use of any single module significantly improves the OA for each dataset. However, when either of the two additional comparison modules is added, the OA for each dataset decreases. When both of our proposed modules are used together, compared to not using any modules, the OA for the four datasets increased by 1.25%, 1.34%, 0.35%, and 2.23%, respectively. When two additional comparison modules are added and used together, compared to not using any modules, the OA for the four datasets decreased by 4.86%, 0.41%, 0.65%, and 2.91%, respectively. Notably, OA can be enhanced when the added comparison module adds any of the modules we have proposed individually, indicating that each of our proposed modules plays a role in the HSI classification process.

2) DBSSC Module Ablation Study: We conducted an in-depth investigation of the DBSSC module and compared the effects of the SpaC module, the SpeC module, and the DBSSC module on classification performance. The rest of the network architecture remained consistent. Table XI displays the OA classification results of the three convolution modules across four datasets. In most datasets, the spectral-based convolution model outperforms the spatial-based convolution model. It can be seen that spectral features play a very important role in HSI classification. Furthermore, across all datasets, the DBSSC model performs better than other convolution models. This indicates that the DBSSC module integrates multiscale spatial and spectral information, further enhancing the capability of shallow feature extraction in HSIs.

3) GMSA Module Ablation Study: To further validate the effectiveness of the GMSA module, we compared its impact with that of the original self-attention (OSA) mechanism and our proposed MWSA on classification performance. Before performing attention calculations, we also enhanced global features. To maintain fairness, we applied GFE to both types of attention mechanisms and then compared their performances again. Table XII displays the OA classification results across four datasets. It can be observed that, compared to the OSA, our method shows improvements of 1.33%, 1.85%, 0.14%, and 0.94% on the four datasets, respectively. Even with the addition of GFE, our approach maintains its superiority. This is due to our proposed MWSA, which considers local spatial information in deep feature extraction. Additionally, after

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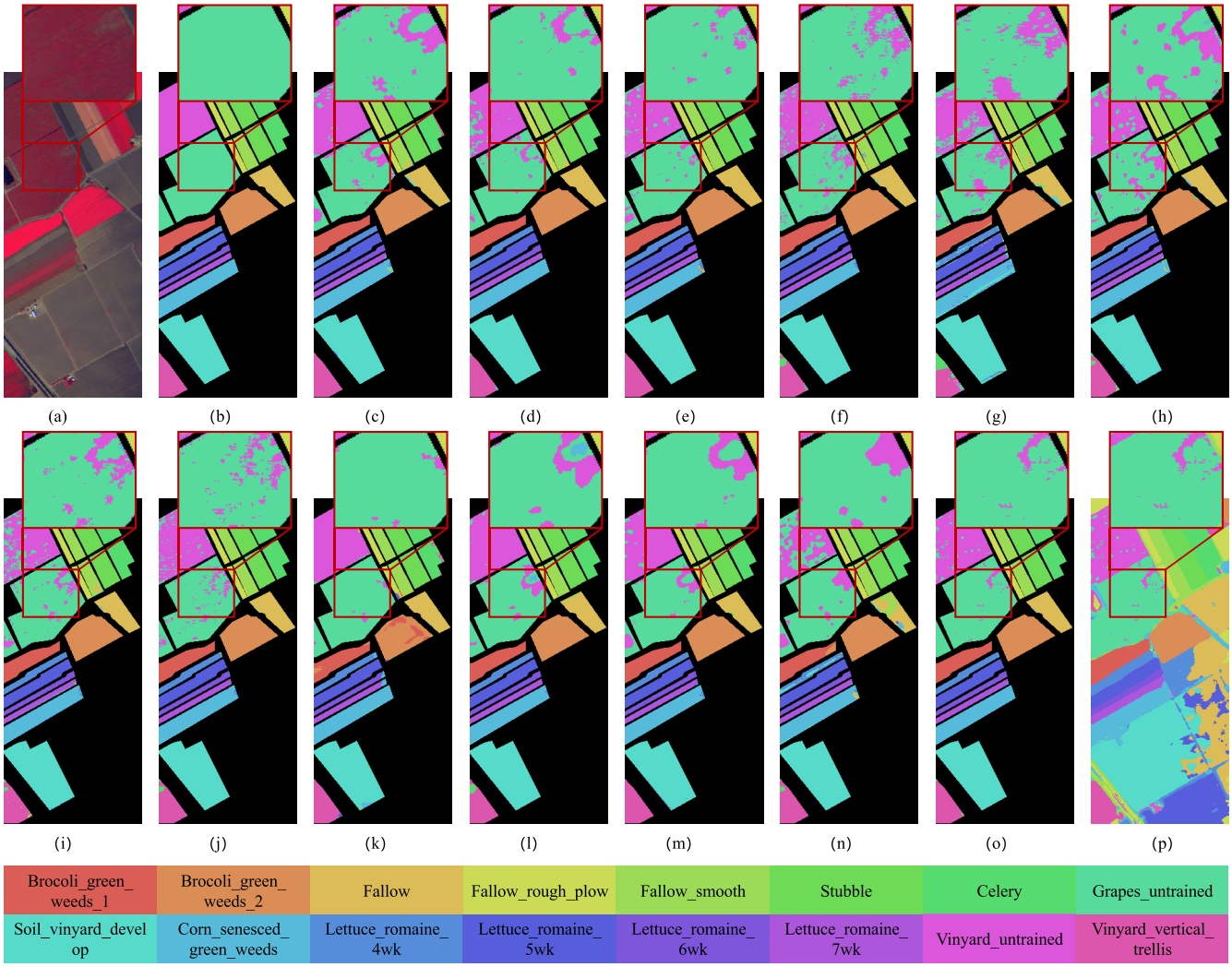


Fig. 5. Classification visualization plots for all methods on the SA dataset. (a) False color image. (b) Ground truth. (c) HybridSN. (d) SSRN. (e) DBDA. (f) RSSAN. (g) ViT. (h) SpectralFormer. (i) SSFTT. (j) GAHT. (k) SSWinT. (l) MVAHN. (m) morphFormer. (n) MHCFormer. (o) H2MWSTNet. (p) All pixel classification map by H2MWSTNet.

enhancing the input features globally, we effectively mined key information, significantly boosting the features in the Q, K, and V feature maps.

4) LFE and LEPE Module Ablation Study: To evaluate the roles of the LFE and LEPE modules in the MWSFormer encoder block, we designed and conducted a series of ablation experiments. By gradually removing these two modules and observing the impact on the model's classification performance, we aim to validate their importance in HSI classification. The results are presented in Table XIII. After removing the LFE module, the classification performance significantly declined, indicating that the LFE module is crucial for capturing local features. After removing the LEPE module, the classification performance decreased, suggesting that the LEPE module plays an important role in providing positional information and helping the model understand spatial structure. When both the LFE and LEPE modules were removed, the classification performance further declined, confirming the importance of the synergistic effect of these two modules in enhancing the model's performance.

5) Hierarchical Structure Ablation Study: To explore the influence of the hierarchical structure on the proposed model, we compared the hierarchical and nonhierarchical versions of H2MWSTNet, with results shown in Table XIV. Their network models are the same, except that the nonhierarchical network models do not perform layer-by-layer channel decrementing in the MWSFormer encoder block, i.e., their number of channels is fixed in each layer. It can be observed that H2MWSTNet with hierarchical structure achieves higher classification accuracy with fewer parameters.

E. Comparative Experiments

1) Quantitative Analysis: We conducted extensive experiments on four datasets, including the SA dataset, PU dataset, WHU-LK dataset, and HR-L dataset, to quantitatively analyze performance metrics such as OA, AA, and κ. All results from our experiments are listed in the form of mean $\pm$ standard deviation, and the best results in each row are highlighted in bold, as shown in Tables XV–XVIII. It can be

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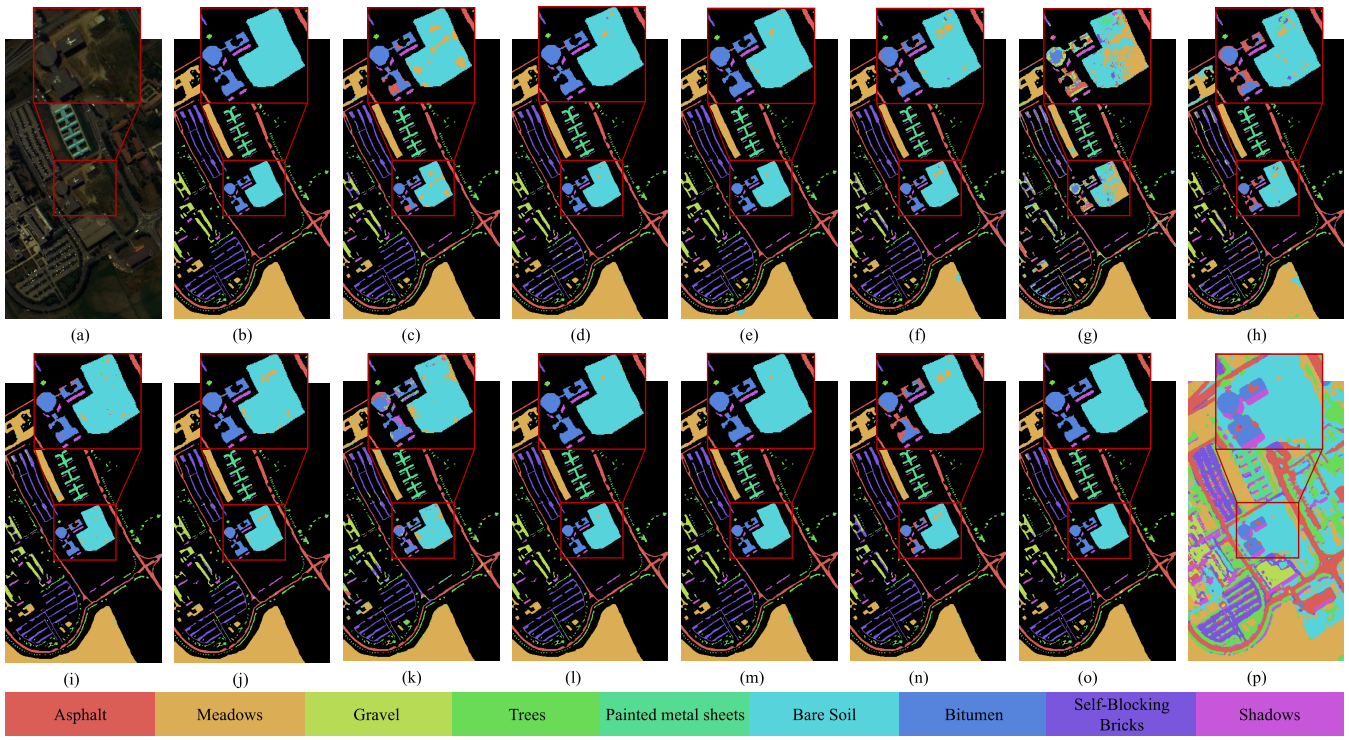


Fig. 6. Classification visualization plots for all methods on the PU dataset. (a) False color image. (b) Ground truth. (c) HybridSN. (d) SSRN. (e) DBDA. (f) RSSAN. (g) ViT. (h) SpectralFormer. (i) SSFTT. (j) GAHT. (k) SSWinT. (l) MVAHN. (m) morphFormer. (n) MHCFormer. (o) H2MWSTNet. (p) All pixel classification map by H2MWSTNet.

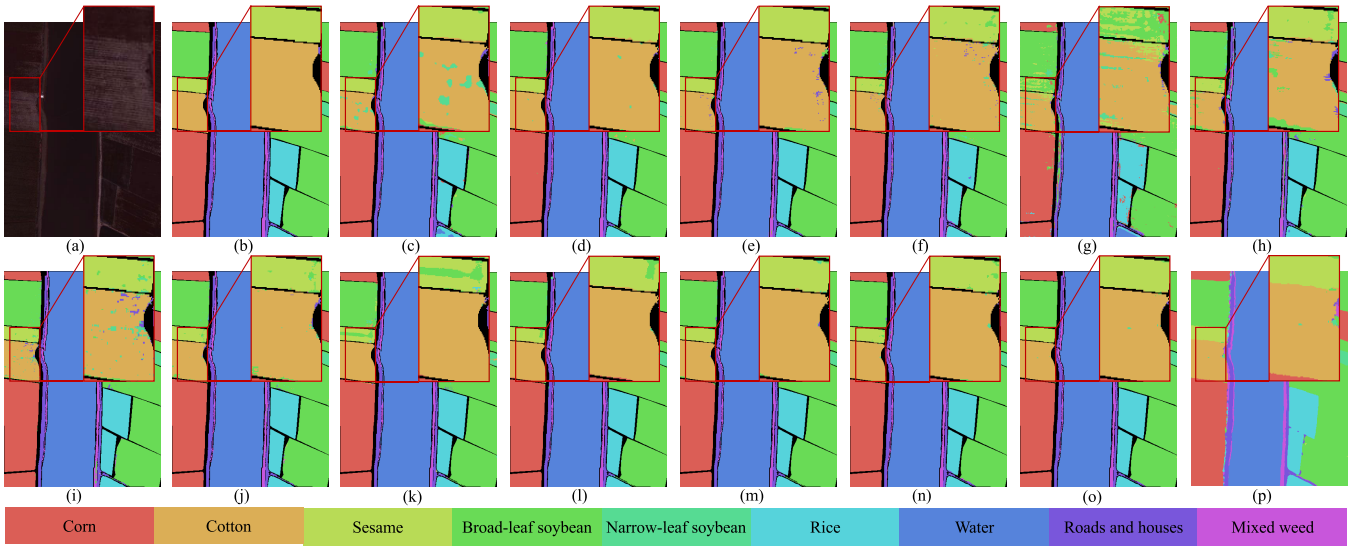


Fig. 7. Classification visualization plots for all methods on the WHU-LK dataset. (a) False color image. (b) Ground truth. (c) HybridSN. (d) SSRN. (e) DBDA. (f) RSSAN. (g) ViT. (h) SpectralFormer. (i) SSFTT. (j) GAHT. (k) SSWinT. (l) MVAHN. (m) morphFormer. (n) MHCFormer. (o) H2MWSTNet. (p) All pixel classification map by H2MWSTNet.

observed that, compared to the most classical and advanced methods, our proposed H2MWSTNet model achieved the best results on all four datasets and exhibited the lowest standard deviation. Moreover, most hybrid CNN–Transformer methods outperform those based solely on CNN, CNN with Attention, and Transformer. This is because the hybrid CNN–Transformer approach leverages the strengths of both CNN and Transformer architectures, maximizing their respective advantages to enhance the model's expressive capabilities. Although both CNNs and Transformers possess significant feature extraction capabilities, each standalone network model has its limitations. Only by combining the two can there be a complementary extraction of rich spatial and spectral information. Furthermore, compared to methods that employ only CNN or Transformer architectures, CNN with Attention achieves better results. This is because the CNN with Attention enhances the model's feature representation capabilities by incorporating the Attention mechanism, which aids in handling long-range dependencies, reduces information loss, and adapts more effectively to varying data distributions,

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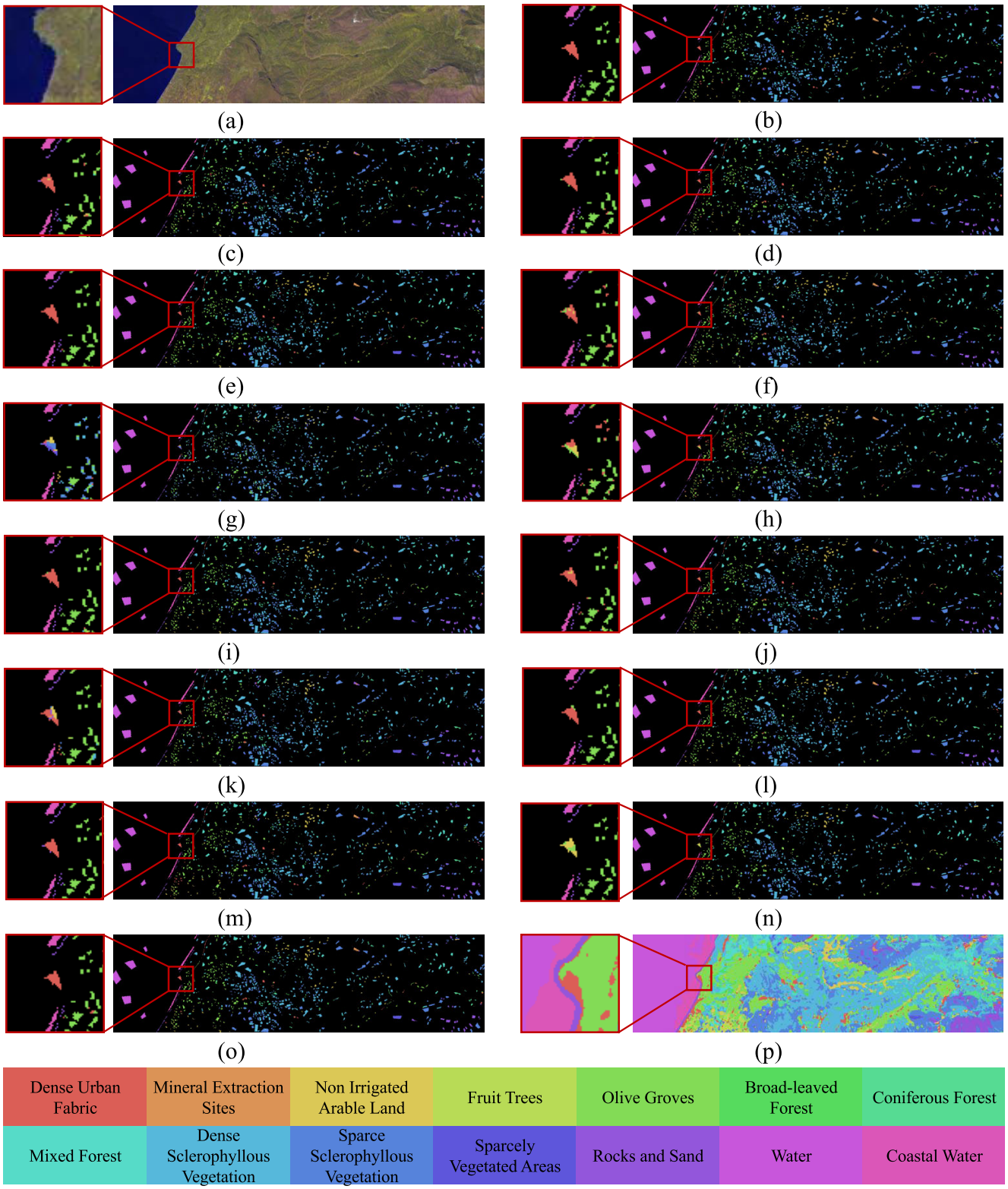


Fig. 8. Classification visualization plots for all methods on the HR-L dataset. (a) False color image. (b) Ground truth. (c) HybridSN. (d) SSRN. (e) DBDA. (f) RSSAN. (g) ViT. (h) SpectralFormer. (i) SSFTT. (j) GAHT. (k) SSWinT. (l) MVAHN. (m) morphFormer. (n) MHCFormer. (o) H2MWSTNet. (p) All pixel classification map by H2MWSTNet.

thereby demonstrating superior performance in HSI classification tasks. It is noteworthy that, to better evaluate the performance and stability of the model, we used the average of five experimental runs as the result.

Table XV lists the classification results for the SA dataset. H2MWSTNet performs better in most land cover categories; however, for certain categories, the accuracy of other algorithms is close to or even surpasses that of H2MWSTNet. This

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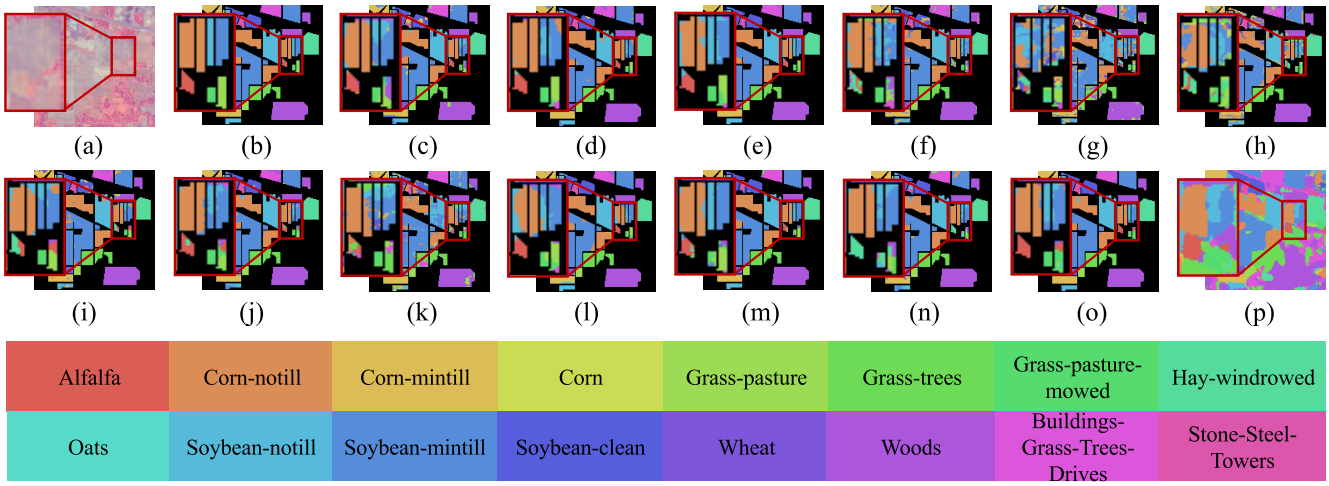


Fig. 9. Classification visualization plots for all methods on the IP dataset. (a) False color image. (b) Ground truth. (c) HybridSN. (d) SSRN. (e) DBDA. (f) RSSAN. (g) ViT. (h) SpectralFormer. (i) SSFTT. (j) GAHT. (k) SSWinT. (l) MVAHN. (m) morphFormer. (n) MHCFormer. (o) H2MWSTNet. (p) All pixel classification map by H2MWSTNet.

is due to the imbalanced sample distribution within the SA dataset, resulting in fewer samples for some categories. Among all algorithm models, the categories "Grapes_untrained" and "Vinyard_untrained" exhibit relatively poor classification performance. This is due to the highly similar spectral features and contours inherent to these categories, leading to misclassifications that impact overall performance. However, H2MWSTNet still maintains the highest OA, AA, and κ. Compared to the GAHT model, it shows improvements of 1.25%, 0.64%, and 1.39%, respectively. Although both models are hybrid CNN–Transformer architectures, our approach emphasizes multiscale spatial and spectral information at the shallow levels and considers local spatial information at the deeper levels, making the extracted features more discriminating and significantly boosting the representational power of H2MWSTNet.

In the experiments conducted in Table XVI, it can be observed that H2MWSTNet achieves the highest classification accuracy in the four categories of the PU dataset: "Gravel," "Trees," "Painted metal sheets," and "Self-Blocking Bricks," surpassing the results of other comparative algorithms. Among Transformer-based comparative methods, ViT achieves the lowest accuracy for the "Painted metal sheets" category, whereas H2MWSTNet reaches 100% accuracy in identifying this category. Additionally, three other algorithmic models also achieve 100% accuracy in recognizing this category. For the DBDA method, although its OA, AA, and κ are very close to our method, H2MWSTNet achieves the highest accuracy in the classification of these four individual categories. Compared to DBDA, our method's OA, AA, and κ values are respectively 0.35%, 0.35%, and 0.46% higher.

In the WHU-LK dataset, the categories "Broad-leaf Soybean" and "Narrow-leaf Soybean" have similar spectral bands. In distinguishing these categories, H2MWSTNet outperforms most comparative methods. In Table XVII, MVAHN, morphFormer, and MHCFormer perform worse by 0.14%, 1%, and 3.53%, respectively, compared to H2MWSTNet in identifying the "Narrow-leaf Soybean" category. Thus, our method excels at distinguishing highly similar samples and attains the highest scores across three evaluation metrics.

HR-L is a highly challenging dataset. In Table XVIII, the "Water" category achieves a classification accuracy of 100% across all algorithms. This is because the HR-L dataset is relatively small and independent, and water bodies have distinct characteristics in the HR-L dataset, including low reflectivity and consistent spectral properties. This makes it easy for algorithms to distinguish between water bodies and other land cover categories. Although the "Coastal Water" category does not perform as well as the "Water" category, five comparative algorithms still achieve a classification accuracy of 100%. Despite the subpar performance of all comparative algorithms on the HR-L dataset, our proposed method, H2MWSTNet, remains ahead of other methods. The evaluation metrics of our method, OA, AA, and κ, are 85.12%, 80.43%, and 82.29%, respectively.

Moreover, on some classic datasets, such as the IP and HT datasets, H2MWSTNet still performs excellently. Specifically, in Table XIX, our method achieves the highest values across all three evaluation metrics. In Table XX, our method attains the best results in almost all categories, with four categories achieving perfect results.

We have demonstrated that the proposed method outperforms other methods across different datasets, particularly in handling multiscale spatial–spectral information and deep local spatial information, resulting in better classification accuracy. However, at the same time, we recognize some limitations of the method. For example, in Table XV, the accuracy for the "Soil_vinyard_develop" class is slightly lower than that of SSRN. This may be due to the high complexity of the spatial distribution or spectral features of this class, leading to some information loss in H2MWSTNet when fusing multiscale information. In Table XVI, the accuracy of our method for the "Bitumen" class is slightly inferior to that of MVAHN. This may be because MVAHN has a stronger capability to capture local spatial information in its specific network structure or has better optimized this class through the attention mechanism. In Table XVIII, more than half of the classes are not at optimal accuracy, slightly lagging behind other methods. This could be due to the presence of imbalanced samples in some categories within the dataset,

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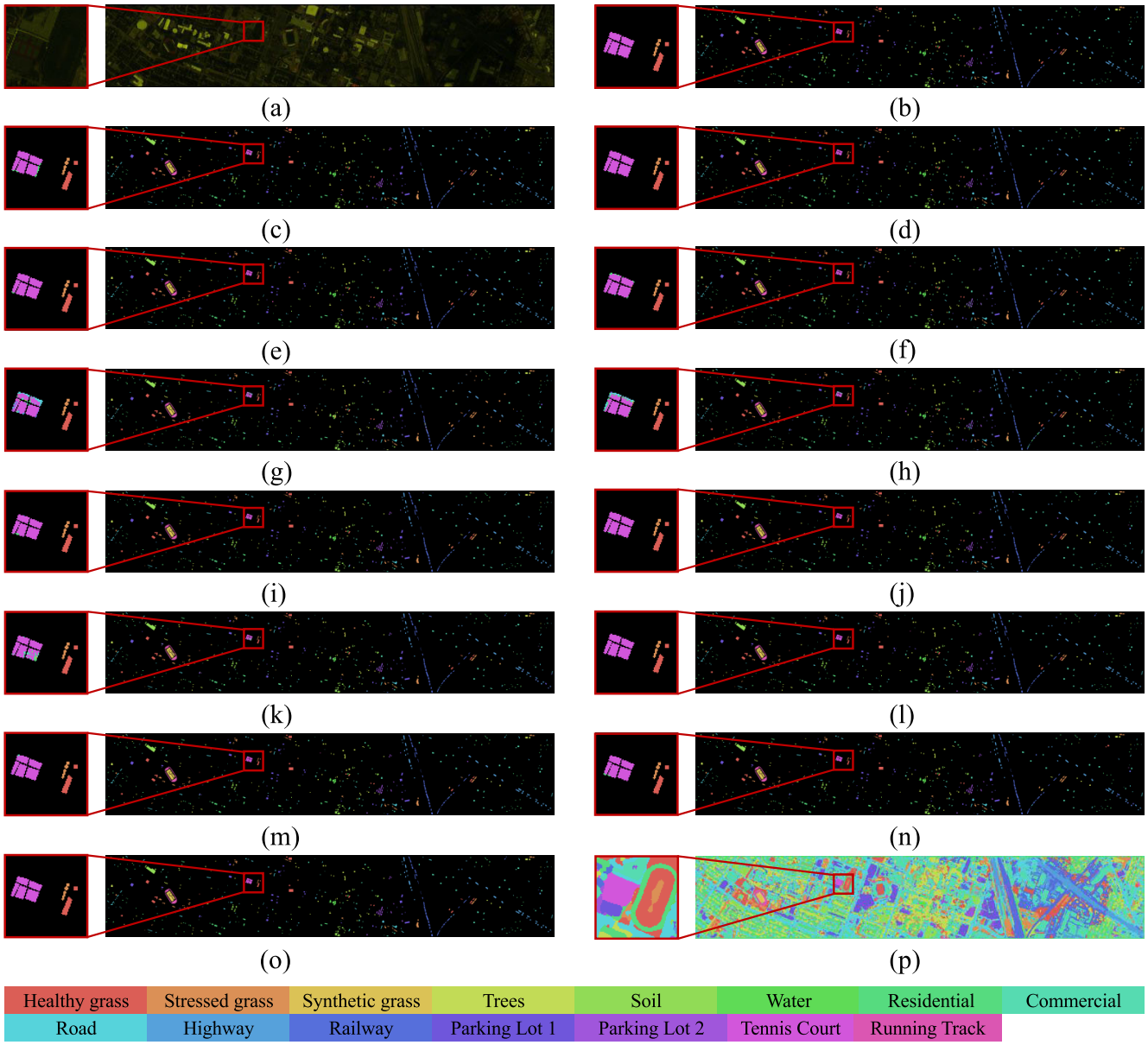


Fig. 10. Classification visualization plots for all methods on the HT dataset. (a) False color image. (b) Ground truth. (c) HybridSN. (d) SSRN. (e) DBDA. (f) RSSAN. (g) ViT. (h) SpectralFormer. (i) SSFTT. (j) GAHT. (k) SSWinT. (l) MVAHN. (m) morphFormer. (n) MHCFormer. (o) H2MWSTNet. (p) All pixel classification map by H2MWSTNet.

resulting in weaker generalization of H2MWSTNet during training for these categories.

2) Qualitative Analysis: To facilitate a more intuitive comparison of different models, we qualitatively analyzed the classification result maps of 12 algorithms across four datasets. As shown in Figs. 5–10, different colors represent different land cover categories. From the visualizations, we can see that our method, H2MWSTNet, exhibits superior performance across all four datasets, with minimal noise, and provides accurate classification results overall. In contrast, both CNN-based and Transformer-based classification algorithms introduce too much noise.

On the SA dataset, most comparative methods perform poorly in the pink area at the top left corner and the green area in the middle, with a lot of noise. Our method, in contrast, has relatively little noise. For the PU and WHU-LK datasets, our method appears closer to the ground truth on some boundaries and major categories. For the HR-L dataset, areas with blue and red spots in the image are challenging to identify, and almost all comparative algorithms struggle to accurately delineate their contours and surrounding regions. Our algorithm largely restores the contours and the areas of the red and blue spots, which further validates the effectiveness of our method in terms of discrimination performance.

For the IP and HT datasets, our method demonstrates excellent performance in terms of region consistency, noise suppression, and category differentiation accuracy, showing a clear advantage in the classification tasks involving complex land cover.

3) Complexity Analysis: We evaluated the complexity of various methods on four datasets, including floating point operations (FLOPs), number of parameters (Params), and the average training and testing time over five runs, as shown in Table XXI. Notably, our proposed H2MWSTNet method

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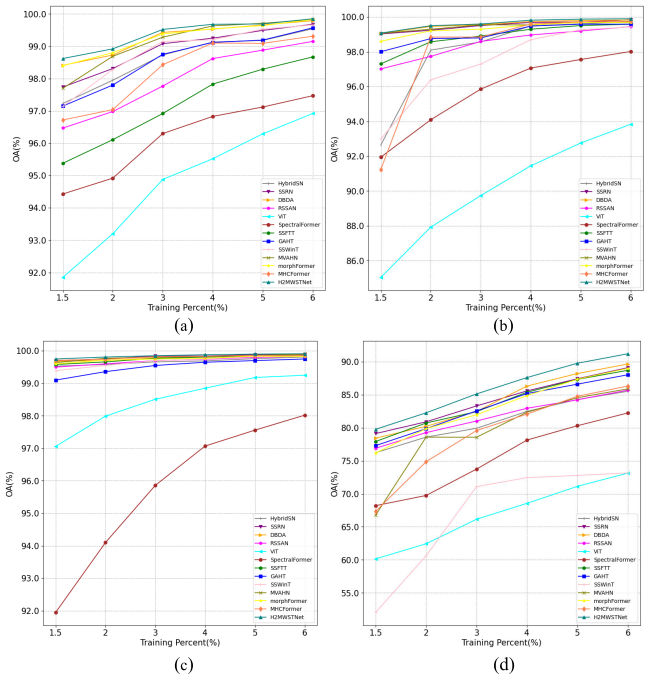


Fig. 11. OA (%) under different percentages of training samples on the four datasets. (a) SA. (b) PU. (c) WHU-LK. (d) HR-L.

demonstrates commendable performance. However, it must be acknowledged that there is still room for improvement in terms of speed and efficiency, which also presents a limitation of the H2MWSTNet method. Regarding time complexity, HybridSN exhibits longer training times on each dataset, particularly on the WHU-LK dataset, where its training time reaches 50.27 min, significantly exceeding that of other methods. RSSAN and SSFTT show high efficiency with shorter training and testing times on each dataset. This discrepancy may be due to the large amount of 3-D and 2-D convolutional computational effort imposed by the HybridSN method in pursuit of robust feature representation. Regarding computational complexity, our H2MWSTNet method has moderate parameters, and the FLOPs are at a medium level compared to all other methods. For example, on the SA dataset, the FLOPs for ViT and MHCFormer are 655.83 and 322.74 M, respectively, while the FLOPs for H2MWSTNet are 184.29 M. It must be emphasized that although H2MWSTNet may not be the most efficient method compared to others, it consistently achieves state-of-the-art accuracy across all datasets, as shown in Tables XV–XVIII.

4) Robustness Analysis: To verify the robustness of H2MWSTNet, we randomly selected different numbers of training samples from the four training sets and used different comparison methods as our competitors. Specifically, we set the training sample sizes to 1.5%, 2%, 3%, 4%, 5%, and 6%, conducting each experiment five times. As shown in Fig. 11, the OA results for the 13 methods across various dataset sizes are displayed. We observe that as the percentage of training samples increases, the OA results gradually improve. When the sample sizes of datasets such as SA, PU, and WHU-LK reach a certain number, the classification results of most algorithms tend to stabilize, while in the HR-L dataset, the results still show an upward trend. Furthermore, our proposed method significantly outperforms other competitors and demonstrates robustness. This further indicates that the exceptional performance of H2MWSTNet may be due to our effective exploitation of multiscale spatial–spectral information and local spatial information in CNN and Transformer.

V. Conclusion

In this work, we introduced a hybrid hierarchical MWSFormer method for HSI classification, termed H2MWSTNet. This method employs a hierarchical structure, effectively reducing information redundancy by progressively decreasing the channels layer by layer. It utilizes a continuous local–global strategy overall, modeling low, medium, and high-level features separately. We first designed spatial and spectral branches using 2D-CNN to extract more discriminative features in the spatial and spectral domains, respectively. Then the feature information from these two branches is summed up by feature fusion to obtain the shallow, multiscale spatial–spectral semantic information of HSIs. Next, to capture spatially sensitive features deep within the model, we input the shallow semantic information into an MWSFormer encoder based on GMSA. This reduces dependency on positional information and better captures the semantic correlations between different bands in local sequences, effectively extracting global spectral and local spatial information. Finally, classification is performed using GAP and a linear layer. Extensive experiments demonstrate that our method surpasses other advanced and classical methods, exhibiting strong generalization capabilities.

In our future work, we will try to extract richer spatial–spectral features from multiple visual perspectives, such as 3D-CNN and GCN. Next, we plan to lighten the design of the Transformer model from the perspective of linear attention to reduce the parameters and decrease the computational complexity. Additionally, we will explore how to fully utilize HSI information with a small sample size, such as through unsupervised learning, self-supervised learning, and transfer learning.

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